TO ESTABLISH A SMALL SCALE INDUSTRY FOR MANUFACTURING OF

"A MANUNAL HUDRAULIC JACK WITH RETRACTABLE HANDLE"

This project reports is being submitted in partial fulfilment of the requirements for the award of

DIPLOMA IN ENGINEERING

(MECANICAL ENGINEERING)

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2023-24

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Certificate

It is to certify that the project entitled "To establish a small Scale industry for manufacturing of the manual hydraulic jack with retractable handle" Submitted by AYAZ IRFAN, MANISH KUMAR SHARMA, HARSHIT VASHISHTHA, SHAHZEB AHMAD, ABDUL MOMIN, SHARAD KUMAR, AASMOHAMAAD, ZUHAIB AHMAD, DEVANSH AGRAWAL of the requirement in partial fulfilment of the award of Diploma Engineering (Mechanical Engineering) in a bona fide record of work carried out by them under my supervision.

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ACKNOWLEDGEMENT

I am very much indebted to my respected supervisor Mr. Shamshad Ali of University Polytechnic, AMU, Aligarh whose motivation, expert advice, vigilant guidance, healthy criticism and affectionate attitude during various stages of this work helped me immensely in completing my project work.

I thank my colleagues AYAZ IRFAN, MANISH KUMAR SHARMA, HARSHIT VASHISHTHA, SHAHZEB AHMAD, ABDUL MOMIN, SHARAD KUMAR, AASMOHAMAAD, ZUHAIB AHMAD, DEVANSH AGRAWAL I would like to express my profound thanks to DR. MOHD ASIF HASAN (Associate Professor) Section In charge Mechanical Engineering Section, and Prof. Arshad Umar, Principal, University Polytechnic, A.M.U., Aligarh for providing necessary facilities and moral support.

Moreover, I express my sincere appreciation to staff of this section who helped me in my project work. I am highly indebted to my great parents and my family for giving me moral support and encouragement.

CONTENTS

CERTIFICATE …….. 2

ACKNOWLEDGEMENT…….. 3

S.NO.	*CHAPTER*	*PAGE NO.*
1.	*INTRODUCTION*	
2.	*DEMAND AND MARKETIBILTY*	
3.	*SITE SELECTION*	
4.	*ORGANIZATION DESIGN*	
5.	*SOURCE OF FINANCE*	
6.	*INSPECTION AND QUALITY CONTROL*	
7.	*WAGES AND INCENTIVES*	
8.	*PROCESS PLANNING AND WORK PLACE LAYOUT*	
9.	*COST ANALYSIS*	
10.	*PRODUCT DESING AND ASSEMBLY DRAWING*	
11.	*RAW MATERIAL*	
12.	*FINANCIAL ESTIMATE*	
13.	*BREAKE EVEN ANALYSIS*	

14.	*DISCUSSION AND CONCLUSION*	

CHAPTER -1

INTRODUCTION

INTRODUCTION

What is jack:-

A jack is a mechanical lifting device used to apply great forces or lift heavy loads. A mechanical jack employs a screw thread for lifting heavy equipment. A hydraulic jack uses hydraulic power. The most common form is a car jack, floor jack or garage jack, which lifts vehicles so that maintenance can be performed. Jacks are usually rated for a maximum lifting capacity (for example, 2 tons or 4 tons). Industrial jacks can be rated for many tons of load.

Followings are the types of lifting jacks:-

- Hydraulic jack.
- Bottle Jacks.
- House Jacks.
- Inflatable Jacks.
- Ratchet Jacks.
- Scissor Jacks.

- Screw Jacks.
- Service or Floor Jacks.
- Transmission Jacks.

HYDRAULIC JACK:-



Hydraulic jacks are mechanical devices used to lift heavyweight. They are commonly used in automobile industries to lift vehicles from the ground, which is why they are known as a kind of hydraulic car jack. Normally, jacks are rated for a maximum lifting capacity, for example, 1.5 tons or 3 tons. Industrial jacks can be rated for many tons of load.

The force of hydraulic jacks can be calculated with pascal's law. Hydraulic jacks consist of two pistons inside the cylinder that is responsible for lifting the heavyweight. The bigger and smaller cylinder are connected together. The small piston forces the fluid to lift the big piston. There are two ball valves that work as a check valve, allowing flow and stop the flow while pumping.

Pressure = Force X Area

A bar is inserted into the handle socket of the jack, when pumped, the hydraulic fluid forces by a piston and go through a one-way valve. The fluid force acts at the bottom of the bigger cylinder, allowing the ram to go upwards. The other valve is blocked so pressurized fluid can't back until it is unscrewed. This is how the hydraulic jack works.

Applications of hydraulic jacks:-

Jacks as a whole have wide applications in different industries and are used to lift and transport heavy loads, relieve vertical forces from weight-bearing members to create space for repairs. Hydraulic jacks or other kinds are also equipment used in construction work, shipbuilding, automotive repair, and other specialized industries. Below are the applications of all types of jacks including hydraulic jacks.

- Automotive
- Bridges
- Cable Aircraft
- tensioning
- House moving
- Construction
- General industrial
- Laboratory
- Locomotive
- Marine
- Mobile home

- Railroad
- Shipbuilding
- Shoring and stabilizing such as for tunnels and mines
- Tractors & trailers

In essence, the applications of hydraulic jacks include;

Lifting industrial heavy loads.

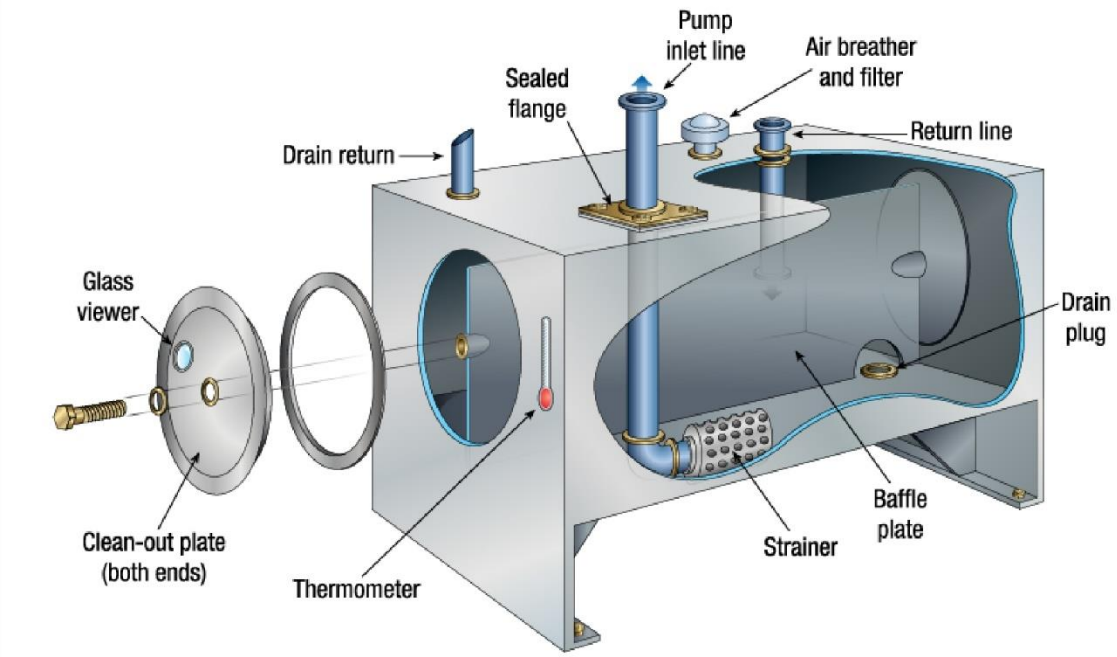
- Lifting cars and vehicles for clamping.
- Finally, clamping machinery equipment.

Components of hydraulic jacks

illustrate and understand the components of hydraulic jacks, a typical manually operated hydraulic jack will be considered since it's commonly used. Below are the major components of hydraulic jacks.

- Reservoir or buffer tank
- Pump with piston or plunger
- Pump lever or handle
- Check valve
- Main cylinder or ram
- Release valve
- Handle

A reservoir or buffer tank – is a vessel that holds the hydraulic fluid or pumps oil which will be used to transfer applied pressure from the pump to the ram.



Piston pump –Piston pump is a part that is mechanically activated by moving the pump lever or handle up and down.



Pump lever or handle – the movement of the pump lever builds up pressure in the hydraulic fluid which transfers that pressurized fluid through a check valve and into the main cylinder.



The main cylinder –The main cylinder is also referred to as ram, when drive upwards, it extends out of the hydraulic jack



Pressure by the body hydraulic fluid, creating the required lifting force and lifting the load.

Release valve –It is a part that helps to release. the pressure so that the ram can retract and the load can be lowered



Above is the diagram of hydraulic jacks, including the small components:



1. Inner Handle

2. Outer Handle
3. Alligator Clip
4. Pins
5. Pump Connector Rod
6. Link Connector Screw
7. Pump Handle Housing
8. Pump Rod
9. Pump O-Ring
10. Pump Plastic Ring
11. Pump Housing
12. Pump Housing Washer
13. Valve Screw
14. Valve Screw O-Ring
15. Valve Ball Bearing
16. Base
17. Plastic Bumper
18. Plastic Fluid Filter
19. Base Plastic Washer
20. Ramrod Rubber O-Ring
21. Ramrod Plastic O-Ring Retainer
22. Ramrod Base
23. Ramrod Shell
24. Extender Screw
25. Inner Chamber
26. Outer Cylinder
27. Rubber Plug
28. Large Plastic Cap Ring
29. Outer Cylinder Screw Cap

Working principle of hydraulic jack

The working of hydraulic jacks is less complex and can be easily understood. The above explain type of jack was the bottle type. Watch the video below to learn how other types of hydraulic jack works:

Advantages and disadvantages of hydraulic jacks

Advantages:

Below are the benefits of hydraulic jacks in their various applications.

- Most designs occupy less space.
- Less likely to jam due to rust in the screw thread.
- Highly effective with heavy loads.
- It is easier to use.
- Loads are lifted with minimum effort.
- It is lighter than screw jacks.

Disadvantages of old hydraulic jack

Despite the good advantages of hydraulic jacks, some limitations still occur. Below are the disadvantages of hydraulic jacks in their various applications.

- It has a relatively low speed.
- Susceptible to failure if oil seals are worn out.
- The device can emit an unpleasant order due to the overuse of warm oil.
- Overheat can quickly occur.
- The leakage of hydraulic oil can cause soil and water pollution.
- The effort of manual power in one working stroke is double.
- Hence it consume more time and energy.

There are main disadvantages of this hydraulic jack

Advantages of new hydraulic jack:-

- The effort of manual power is reduce i.e. only one time effort is used and it comeback in its original position by means of torsional spring.
- Hence it also save time and manual effort.

CHAPTER-2

DEMAND AND MARKETABILITY

INTRODUCTION:

For a person to act setup an industry, its market value or in other sense the reaction of product in market is a great necessity. The market consumes different types of production products and to launch a new product in the market means to setup a new era for the competition of products that are still prevailing in the market.

DEMAND:

The demand of a commodity is essentially consumer's attitude and reaction towards that commodity. Demand for a good is fact a photograph or a panoramic picture of a consumer's attitude towards a commodity. This consumer's altitude gives rise or many give rise to actions in purchasing so many units of a commodity at various given prices. Precisely stated, the demand for a commodity is the amount of it that a consumer will purchase or will be ready to take off from the market at various given prices at a given moment of time. The larger demand for a firm's product, the higher is the price it can charge and thus making profit accordingly

TYPES OF DEMAND:

These are some important ways of categories demand:

1. Demand for consumer's goods and producer's goods.
2. Demand for perishable and durable goods.
3. Derived and autonomous demand.
4. Firm and industry demand.
5. Demands by tool market and market segment.

OBJECTIVES OF DEMAND ANALYSIS:

- To study and analyse the determinants of demand
- To prepare sales or demand forecast
- To measure the elasticity of demand
- Manipulating demand
- To make appropriate changes in allocation of resources

DETERMINATIONS OF DEMAND:

The demand depends on certain factors, the most important of which are as follows:

- Price of the product
- Price of relative goods
- Consumer's income,
- Consumer's taste ft. and preferences,
- Population and its distribution,
- Consumer expectation.

MARKET SURVEY:

Market survey is an analysis of a project to be started, expanded or modified. Broadly speaking market research is the commercial research suitably of a business and is a continuous process, i.e. research is always kept continued for the stability for the stability business. If demand reduces suddenly, production comes to a standstill, which may produce great losses to the. Market research includes forecasting, intelligence and statistics. It is an important product for the sale of the products. Failing to do this accurately may lead to over production. This may result in the financial losses to the firm. It may lead to under production, the result of which is equally unpleasant.

DEFINITION AND CONCEPT:

Market survey may be defined as a way of finding out facts which must be identified before a market policy and has to be determined before starting a business; planning department makes different survey affecting the business to maintain the good working conditions, better quality and have some profits. Many new products and firms fail because they start business as a “GUESS-ESTIMATE”. Therefore, investment of small amount and time a careful market analysis is a must including comparative better knowledge instead of bigger loss at a larger stage.

NECESSITY OF MARKET RESEARCH:

The necessity of a market research existed in 20th century due to industrialization of western societies in 19th century. In earlier days there were fewer gaps between the initial manufacturer and the ultimate consumer. The manufacturer knew his consumer's demands, desires and habits. They were very close to each other because of few in number. But later in the modern period the industry separated their customers by complicated organization and engaged in mass production for cheap and large-scale demands. Therefore, the demand of the customer must be explored, developed and evaluated. The change from intimate manufacturer and customers contact is similar to a broken circular chain.

OBJECTIVE OF MARKET RESEARCH:

Mainly the object of market research is to inform the management ads about to what future held for its resources and proposed products.

The main objective of the market research is to find out:

1. Sales volume.
2. Amount of its profits and its duration.
3. Amount of capital needed.
4. Cost of necessary advertising.
5. Production cost.

6. Stability of raw material supply.

7. Handling and transportation charges.

8. Business condition and trends.

9. Seasonal market characteristics.

FUNCTION OF MARKET RESEARCH:

Any other factors which may be considered relative to the success of the business.

a. It helps in knowing that who ends where the customer is and what he wants.

b. It helps in knowing the sale trend, market which is essential for production planning.

c. It studies the distribution channel and its effectiveness.

d. The exploit new markets and helps developing new products.

e. Its safeguard the interest of the company against changes in the market in future.

f. It keeps the business in touch with its market and thus helps and sales promotion efforts.

CLASSIFCATION OF MARKET RESEARCH:

Market research is sub-divided into four general classifications like described below:

(i) Product analysis,

(ii) Market analysis,

(iii) Distribution analysis,

(iv) Competition analysis.

Product analysis:

Product analysis is required to find out customers preference for the product. This will enable management to make improvement that will meet the requirements of the market and will be more acceptable than those of competitor's products. A check list of questions to be answered by a product analysis is given below:

- Whether the product fulfils market requirements.
- Whether the product competitive in character, performance and price.
- Whether the product permitted efficient distribution.
- Whether the product can be solved at handsome profits.
- Whether the lines complete the required size.
- Whether some design improvement is required to fulfil customer's need and how product can be re-designed to reduce cost of production, packing, distribution,

transportation and maintenance without affecting quality. Whether reduction of a selling price has effects on demand.

- Whether finished stock can be reduced to minimize investment, warehouse cost and obsolescence.
- How repair and replacement methods can be improved?
- What is the future trend of changes as determined by customer's changing habits?
- What new changes and inventions are to occur?
- What is composition offering? Whether some better design, ideas can be taken from competitor productions.

Market analysis:

The objects of market analysis are to be find the location of markets, scope of sales and buying habits of customer that make up the potential markets for a product. Therefore, this fact is required for deciding about the capacity of plan.

A check list of questions to be answered by a markets analysis is given below

1. Who would purchase a product?
2. Where are the markets?
3. How many other customers influence the product?
4. What is the volume of sale during a particular time?

5. On what factors buyers pay stress e.g. quality, price, delivery, safety or service etc.?

6. When are customers mostly in the process of buying?

7. How often they used to purchase?

8. Do they buy in advance of needs?

9. Whether demand is seasonal?

10. What discount and credit terms are prevailing?

Distribution analysis:

This consists of the study of the channels of distribution. Methods of pricing, resale price, maintenance, salary methods, sales promotions sales training, warehouse distribution cost analysis and other policies required for the product distribution. The economic selection of the best channels of distribution for a product is essential for the success in market. It includes transportation, warehousing and inventory costs.

Competition analysis:

Competition studies are generally including as a part of marketing analysis. This makes necessary to study the recent or proposed production sales and pricing policies of the competitors. Most of these are the analysis of the quality of the competitors' product to as certain how a potential customer would compare with ones & own product. From these studies, plans and policies may be made to guide

future activities of a concern in order to improve its competitive position.

MARKET RESEARCH TECHNIQUES:

To carry out various analyses as explained above the following techniques are adopted.

1. Collection of data:

Since these data are on the table of the organization, this technique is also known as desk research. In this method, the data are collected from published material either by the company itself or by other agencies e.g. journals of the trade published survey reports, government publications, international publications etc. This type of research is carried out for past sales, effects of various factors on sales in the past, fluctuation in the sales etc.

2. Interviews:

To know the views of salesman, dealers and consumers interviews are conducted with a selected sample of people. These interviews can be conducted on telephone or in person depending up to the nature of question.

3. Market survey:

In order to collect specific data or views carefully prepared questionnaire is posted to the selected sample of people.

Rate of mild steel- 75 rupees per kg

Rate of spring material (carbon steel)- 95 rupees per kg

Salary of lathe machine operator- 17850 rupees

Salary of drill machine operator- 9996 rupees

Salary of spring maker machine operator- 7980 rupees

4. Statistical Methods:

Statistical methods like bar chart, frequency polygon, distribution curves and concepts of standard deviation are used to serve the purpose.

CHAPTER 3

SITE SELECTION

SITE SELECTION

INTRODUCTION

The process of site selection is a strategic decision that impacts an industry in various ways such as plant layout, capital investment, operating costs, efficiency, productivity, and profitability. It involves two stages: determining a general area or region, and then selecting a specific site within that area.

Let's break down the factors that influence these decisions in more detail:

Raw Material Availability:

The proximity to raw materials is essential, especially if those materials are heavy or bulky. The closer the industry is to the raw materials, the lower the transportation costs, which can significantly impact the overall operating costs.

Market Proximity:

Locating near markets helps in reducing the cost of transporting finished goods. It also allows for quicker

response times to market changes and consumer demands, providing a competitive edge.

Transport Facilities:

Good transport facilities are vital as they ensure smooth and cost-effective movement of raw materials and finished products. This includes access to roads, railways, ports, or airports.

Skilled Labor Availability:

Industries often require a specific type of skilled labour. Hence, the availability of such labour in the region can influence the site selection.

Utility Access:

Access to utilities such as water, electricity, and gas is another crucial factor. Their sufficient supply at reasonable costs is desirable.

Community Facilities:

The presence of community amenities such as parks, shops, cinemas, and educational centres can improve the quality of life for the employees, which can, in turn, result in increased productivity.

Business and Commercial Facilities:

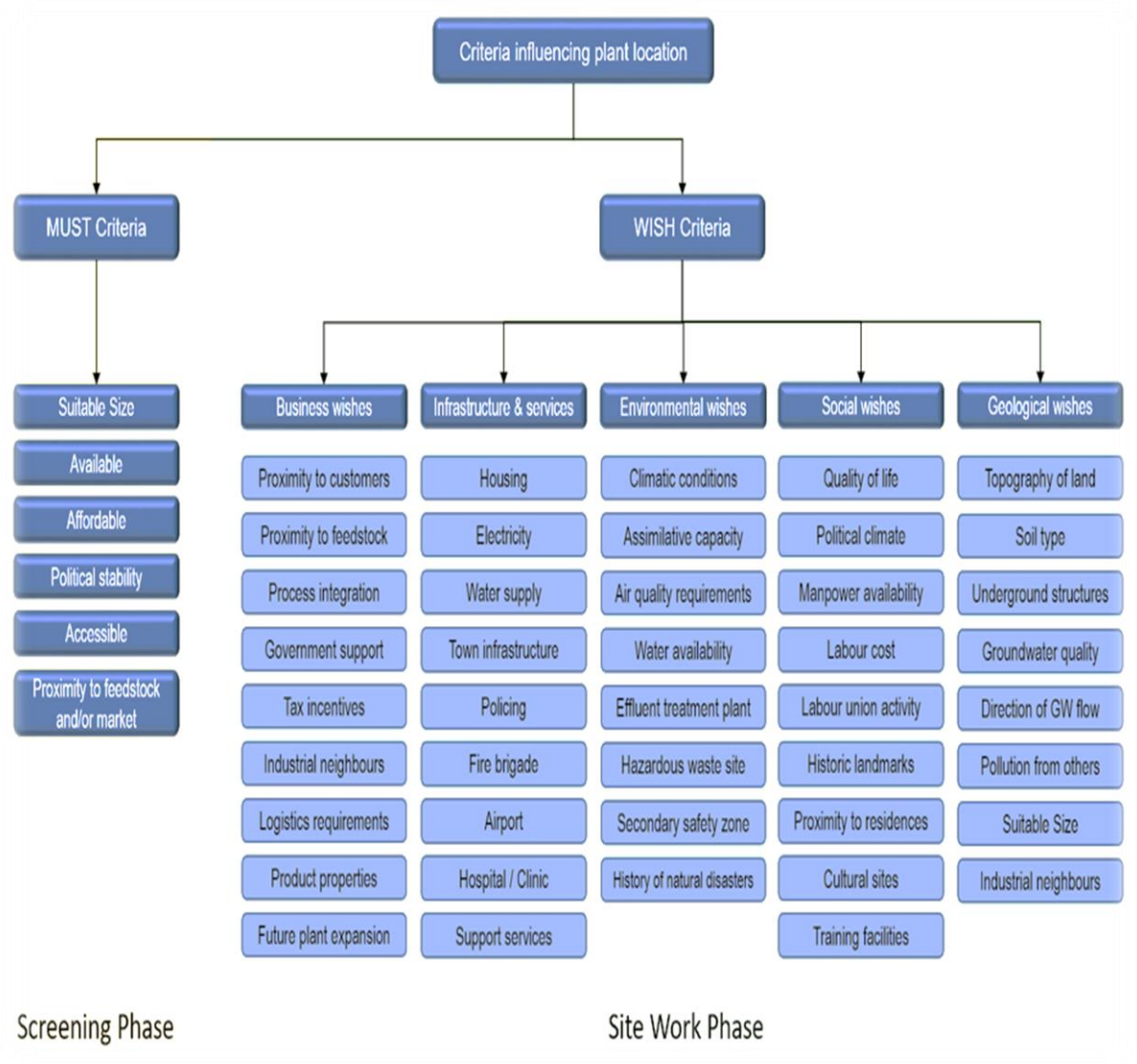
Access to financial and banking services is essential for industries as they require a constant flow of working capital.

Related Industries:

The presence of related industries can be advantageous, especially for small-scale industries. It can facilitate immediate repairs, preventing disruptions in production.

Other Factors:

Factors like local bye-laws, taxes, fire protection facilities, post and telegraph facilities, topography, waste disposal, ecology, and pollution can also influence the site selection.



CRIETERIA OF SITE SELECTION

Availability of Raw-Materials:

The place selected should be such where the raw materials are available easily. There should be an easy approach to the place of raw-materials.

For example:

ADVERTISEMENTS:

(i) Iron and steel industry in Bihar,

(ii) Textile factories in Gujarat and Maharashtra,

(iii) Jute works in Bengal owe their success on account of easy availability of raw-materials. It reduces the cost of transportation.

Selection of Suitable Site Factor *# 2. Nearness to the Market:*

Manufacturing a thing successfully is not sufficient. It is also necessary that the output should find ready market and that the product is sold at a price to yield reasonable profit. This is possible only when the market is not far away. Nearness of the market ensures transportation costless and minimum wastage.

Selection of Suitable Site Factor *# 3. Nearness to Sources of Operating Power:*

ADVERTISEMENTS:

Every industry requires fuel for working the machinery and unless the region has rich fuel resources of power now available are coal, hydro-electricity and oil etc. Coal is the cheapest source of power, but it is very bulky and involves high transportation costs.

Selection of Suitable Site Factor *# 4. Labour Supplies:*

For the successful and un-interrupted working of a factory, availability of adequate supply of labour of the right type at reasonable wages is also very essential. There are some industries in which the inherited skill of the workers in an important factor in the process of manufacturing.

For example—The development of the dying and printing industry in Farukhabad and the glass industry in Ferozabad have been mainly located there due to the availability of skilled labour in these towns.

Selection of Suitable Site Factor

5. Transportation:

Every manufacturing industry requires cheap and efficient means of transportation for the movement of both raw-materials from the source of supply to the factory and finished product from the factory to the markets or the centres of consumption. The location of the plant, should therefore be at a place where adequate transport facilities are available at cheaper rate.

Selection of Suitable Site Factor # 6. Finance:

ADVERTISEMENTS:

No productive activity is possible without the availability of adequate capital. Banks, stock exchanges and other similar institutions help in capital formation and expansion of industry by providing financial help to it from time to time.

Selection of Suitable Site Factor *# 7. Climate:*

Certain industries for their successful working require a special type of climate. For example—Cotton textile industry requires humid climate while the photographic goods industry flourishes best in regions of dry climate. Climate also affects the efficiency of labour.

Selection of Suitable Site Factor *# 8. Momentum of Early Start or Industrial Inertia:*

There is a tendency for an established industry to remain localised in a particular areas in which it arose even after some of the original advantages possessed by that area for such work have lost their previous importance. If however, the entrepreneur acts rationally and his necessary knowledge he will choose the location which offers the lowest cost per unit of output.

Selection of Suitable Site Factor *# 9. Personal Preferences:*

Location of any industry may sometimes be decided according to the personal preferences and prejudices of the industrial enterprises.

Selection of Suitable Site Factor *# 10. Government Policy:*

These days the government plays an important role in determining the location of new industries.

In addition to the factors discussed above cost of land and building for setting up the factory topography of the area; the possibilities of future expansion etc., are some other factors which influence the decision-making regarding location of industry.

CONCLUSION

Once a general area has been chosen based on these factors, the next step is to select a specific site within that area. This decision should consider technical, commercial, and financial aspects to maximize advantages and minimize potential issues. Factors such as community attitude, community facilities, transportation facilities, topography, waste disposal, ecology, and pollution should be evaluated during this process.

In conclusion,
We have decided to set-up our industry in Aligarh.

CHAPTER 4

ORGINISATION DESIGN

ORGINSATION DESIGN

INTRODUCTION

Organization design is a crucial part of managing any enterprise, whether it's a small start-up or a multinational corporation. It involves creating and implementing an effective structure of roles, responsibilities, and relationships that align with the organization's objectives. Let's explore this topic in detail.

First, let's understand what an organization is.

An organization is essentially a **systematic arrangement of people brought together to accomplish certain common objectives**. It's a collective group that works towards achieving a

common goal, and the structure of this group is what we refer to when we talk about 'organization'.

The key characteristics of an organization include:

Group of Persons:

An organization is made up of a group of individuals who work together. These individuals are in a position to communicate with each other and cooperate to achieve the organization's objectives.

Common Objectives:

An organization is created with the purpose of achieving certain common objectives. The members of the organization share a collective goal which drives their actions and decisions.

Cooperative Relationships:

The members of an organization cooperate with each other and work together. The relationships within the organization facilitate the achievement of the common objectives.

Structure of Relationships:

An organization is defined by its structure of relationships. This includes the hierarchical structure, division of labour and the distribution of roles and responsibilities.

Rules and Regulations:

An organization establishes rules and regulations that guide the behaviour and actions of its members. These rules help to maintain order and discipline within the organization.

Function of Management:

Organizing is a key function of management. It involves deciding how best to group organizational activities and resources.

Division of Work:

An organization, work is divided into different parts or processes which are performed by individuals or groups according to their abilities and aptitude. This division of labour contributes to efficiency and specialization.

In essence, an organization is more than just a collection of individuals. It's a structured, dynamic entity that's designed to achieve specific objectives. The design and structure of an organization play a critical role in its ability to function efficiently and effectively.

TYPES OF ORGINISATION DEIGN

Line or Scalar Organization:

Also known as a military organization, it divides business activities into different departments like finance, production, and sales. Each of these departments is further sub-divided into sections.

Functional Organization:

In this type, the difficulties of finding all-round qualified people are overcome by replacing them with specialized individuals. Each supervisor is specialized in a particular field and oversees that specific function across all departments.

Line and Staff Organization:

Ideal for large scale businesses, where managers cannot give attention to every part of management. They have 'staff' to do the work of investigation, research, recording, planning, and advising managers. The line maintains discipline and stability, while the staff provides expert information and helps to improve overall efficiency.

Line, Staff, and Functional Organization:

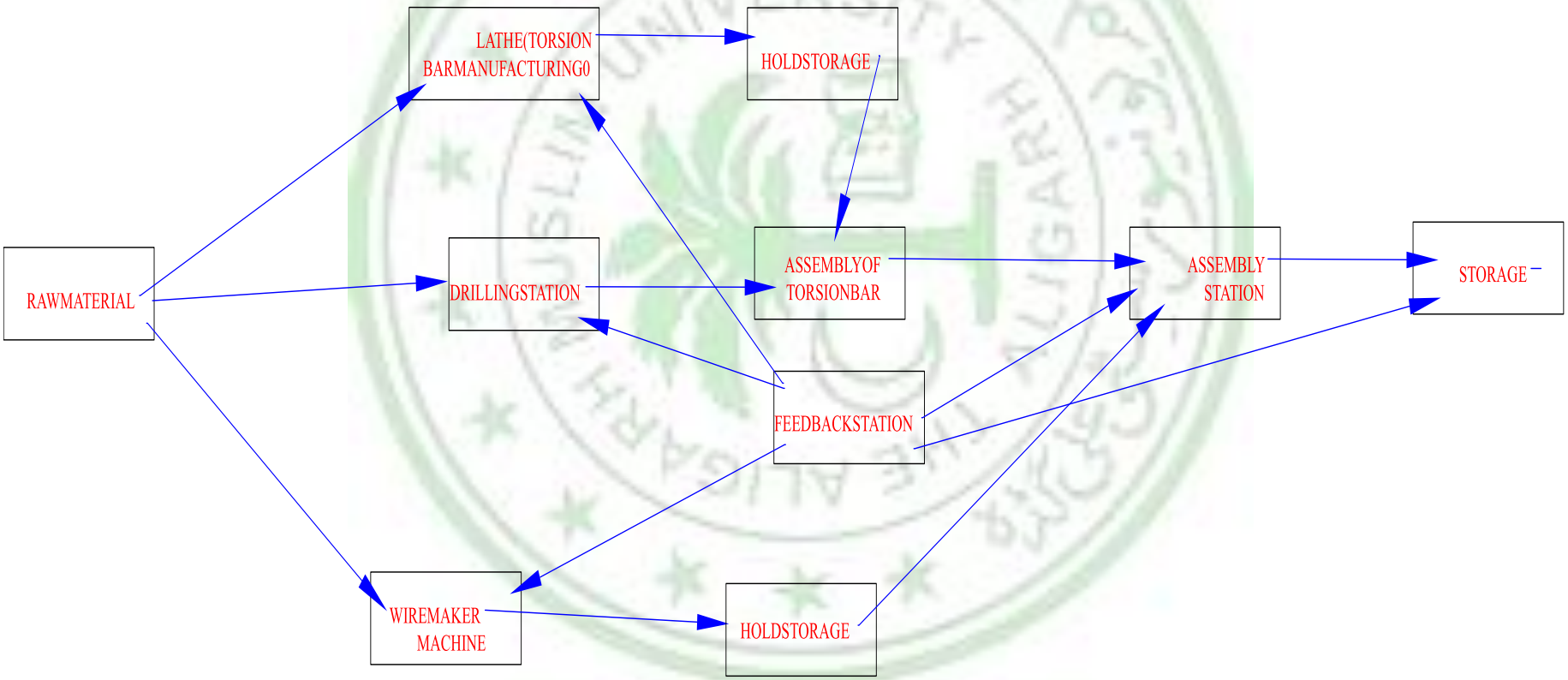
This is a combination of line, staff, and functional organization. It's used to manage discipline and output, with workers kept under the direct control of a foreman.

Committee Organization:

A group of individuals formed to give advice on important problems that cannot usually be solved by an individual alone. Many large companies add a network of committees to their line and staff organization. These committees may be permanent (sometimes referred to as standing committees) or temporary.

These types of organizations each have their own advantages and are applied based on the specific needs and goals of a business.

MANUFACTURINGFLOWDIAGRAM



CHAPTER-5

SOURCE OF FINANCE

The main input that an enterprise or project need to run is finance. Finance of enterprise is all about management of money, how to invest it and other financial assets to achieve Perfection in the product. Finance can be of two types

1- Short term finance
2- Long term finance

Short term finance

It involves the fund required to meet immediate financial requirements and modifications and addresses the short duration needs related to the project

Sources- short term loan, borrowing from the bank or other financial institutions

Long term finance

Long term finance involves obtaining fund for a more extended period usually exceeding one year to support project Main sources of Long term finance are:

1- Loan from bank

2- Equity financing
3- Private equity
4- Crowd funding

Sources for finance

Sources of finance can be broadly categorized into two main types internal and external sources

Internal sources

Internal sources are like finding money from within your own business.
For example

1- Taking loan from relatives or friends
2- Own family savings
3- Loan on your land or jewellery

External sources

External sources of finance involve obtaining funds from outside the business. For example:

1- Taking loan from banks
2- Getting investment from external parties.

In our project we had taken loan of 722000 at interest rate of 7.65% per annum from State Bank of India under Scheme for 10 years.

We have product liability insurance at 5000/month from HDFC ergo for safety of our product and industry. The product liability insurance covers any legal liabilities to third party arising out of injury or harm due to a product sold by a business. It covers the following: Bodily injury, illness or accidental death to a third party. Accidental damage to third party property.

CHAPTER-6

INSPECTION

INTRODUCTION:-

Inspection in the context of project design is a systematic process of reviewing and evaluating design documents, plans, or specifications to ensure their correctness, completeness, and compliance with established standards and requirements. To maintain the accuracy and perfection in the product, special types of gauges and other measuring instruments are used.

Inspection is important for controlling the quality and accuracy in product.

Objectives of inspection

1- Error Detection:

Identify and rectify errors, inconsistencies, or omissions in design documents to prevent issues during project execution.

2 – Quality Improvement: Provide a mechanism for constructive feedback to enhance the overall quality of design deliverables.

3- Continuous Improvement: Contribute to a culture of continuous improvement by learning from identified issues and refining design processes. 4- Cost Reduction:

Minimize the cost of correcting errors by catching them in the design phase rather than during implementation or later stages.

By achieving these objectives, inspection plays a crucial role in promoting the overall success of a project by minimizing risks, ensuring quality

Principles of inspection

1 – Early Detection:

The principle of early detection emphasizes identifying and addressing issues as early as possible in the project life cycle. By catching errors or inconsistencies during the design phase, the likelihood of costly problems during implementation is reduced. Early detection is key to minimizing the impact of issues on subsequent project stages.

2- Systematic Approach:

The principle of a systematic approach involves conducting inspections in a structured and organized manner. This includes following predefined processes, using checklists, and ensuring that the inspection is carried out consistently.

Standards of inspection

1- Inspection of accuracy and precision
2- Inspection of raw material
3- Inspection of completeness
4- Inspection of final finished product

Functions of inspection

The functions of inspection in project design encompass various activities aimed at ensuring the quality, compliance, and accuracy of design documents. Here are key functions:

1- Error detection and correction
2- Quality assurance
3- Efficiency improvement
4- Feedback generation
5- Tools inspection
6- Tool quality inspection
7- Risk mitigation

Methods of inspection

The main three methods of inspection by which we can improve the quality of manufactured products.

1- Visual inspection
2- Dimensional inspection
3- Non destructive testing (NTD)

Visual inspections

This involves visually examining the product or system to identify any visible defects, irregularities, or deviations from specifications. It is one of the simplest and most common inspection methods.

Dimensional inspection

This method focuses on measuring the physical dimensions of a product to ensure they meet specified tolerances and standards. Precision measuring tools are used to assess length, width, height, and other critical dimensions.

Non destructive testing (NTD)

NDT methods allow for the assessment of materials and components without causing damage. Techniques such as ultrasonic testing, magnetic particle testing, and radiographic testing are employed to detect internal defects, discontinuities, or structural issues.

These three methods are fundamental in quality control and assurance across various industries, providing a comprehensive approach to ensuring the integrity and compliance of products or systems.

QUALITY

The quality of a product refers to its ability to consistently meet or exceed customer expectations and comply with specified standards, requirements, or specifications. Key elements of product quality include:

1-Performance: The product should perform its intended function effectively and reliably.

2-Durability: A quality product should have a reasonable lifespan or durability, providing value over time.

3-Reliability: The product's performance should be consistent and dependable under various conditions.

4-Functionality: The features and functions of the product should align with its intended use.

5-Safety: The product should be designed and manufactured with consideration for user safety.

6-Conformance: The product should meet established standards, specifications, or regulatory requirements

7-Ease of Use: A quality product is typically user-friendly and easy to operate.

8-Consistency: Products produced in different batches or units should exhibit consistent quality.

9-Customer Satisfaction: Ultimately, the satisfaction of the end user with the product's performance and features is a crucial measure of its quality.

Factors affecting quality

Several factors can impact the quality of products or services.

Here are some key factors:

1-Raw Materials: The quality of the materials used in production significantly influences the final product. Inconsistent or substandard raw materials can lead to defects or variations in quality.

2-Manufacturing Processes: The methods and procedures used during manufacturing play a crucial role. Well-defined and controlled processes are essential for maintaining consistent quality.

Human Factors: The skills, training, and attentiveness of the workforce impact quality. Adequate training, clear communication, and a culture of quality consciousness are important.

Quality Control and Inspection: The effectiveness of quality control measures, including inspections and testing, is critical. Rigorous quality checks help identify and rectify defects before products reach customers.

Environmental Conditions: Factors such as temperature, humidity, and cleanliness of the production environment can affect quality. Environmental control measures are important, especially in industries with strict quality requirements.

Understanding and managing these factors collectively contribute to the overall quality of products or services.

Functions of quality control department

The Quality Control (QC) department plays a crucial role in ensuring that products or services meet specified quality standards. The key functions of a Quality Control department include:

1-Inspection and Testing: Conducting inspections and tests on raw materials, components, and finished products to verify compliance with quality specifications and standards.

2-Quality Assurance: Implementing processes and systems to prevent defects and deviations from quality standards throughout the entire production

3-Process Monitoring: Continuously monitoring manufacturing or service delivery processes to identify and address any deviations that could impact quality.

4-Documentation and Record Keeping: Maintaining detailed records of quality control activities, including test results, inspections, and corrective actions taken.

The Quality Control department plays a pivotal role in safeguarding and enhancing the overall quality of the organization's offerings.

Indian standard institution

The Indian Standards Institution (ISI) is now known as the Bureau of Indian Standards (BIS). BIS is a national standards body in India, operating under the Ministry of Consumer Affairs, Food & Public Distribution. It was established in 1986 and is headquartered in New Delhi. Some primary features of

ISI are :

1- Development of standard : establishing standards to ensure the quality, safety, and reliability of products and services in India. These standards cover a wide range of industries and sectors.

2- Product Testing: Conducting testing and evaluation of various products to verify their compliance with established standards. This involves rigorous testing procedures to assess the safety and performance of products.

3- Consumer Education and Protection: Educating consumers about the significance of standards and the ISI mark. BIS works to protect the interests of consumers by ensuring that products in the market meet established quality and safety criteria.

These functions collectively contribute to the mission of BIS to enhance the quality of products and services, promote consumer welfare, and support the competitiveness of Indian industries in the global market.

CHAPTER-7

WAGES AND INSENTIVES

Wages

Wages is amount of money is given by employer to the employee like (labour, workers,) for there services .A wage is payment made by an employer to an employee for work done in a specific period of time. Some examples of wage payments include compensatory payments such as minimum wage, prevailing wage, and yearly bonuses, and remunerative payments such as prizes and tip payouts. Wages are part of the expenses that are involved in running a business. It is an obligation to the employee regardless of the profitability of the company. Payment by wage contrasts with salaried work, in which the employer pays an arranged amount at steady intervals (such as a week or month) regardless of hours worked, with commission which conditions pay on individual performance, and with compensation based on the performance of the company as a whole. Waged employees may also receive tips or gratuity paid directly by clients and employee benefits which are non-monetary forms of compensation. Since wage labour is the predominant form of work, the term “wage” sometimes refers to all forms (or all monetary forms) of employee compensation.

Types of wages

- Fixed wages
- Overtime wages
- Time wages
- Cash wages
- Contract wages

Etc.

Fixed wages

A fixed amount of money is paid by the owner or employer to their employees or worker for their services or Fixed pay is the monthly salary companies guarantee their employees in exchange for their services. It includes basic pay and any additional allowances — like housing, childcare, or transport

The amount of fixed pay stays the same regardless of hours worked or individual performance. Some of the main advantages associated with fixed salary are: Employees like the consistency of fixed pay Payroll is simpler to process There's no need to calculate overtime Allowances attract and retain

Overtime wages

When the employee or worker do their work or job to adding extra time to normal time hours.so the employer have

to pay extra amount of money to the workers or employees of extra time services. When you work beyond the standard 40-hour workweek, you may be eligible for overtime pay. While national and local guidelines regulate overtime pay, it may be useful to learn about the general way overtime pay works. Understanding it can help you determine your eligibility and predict how much you can expect to earn when you put in overtime hours at work

Time Wages

The amount of money is paid by the employer or owner to the employee or worker on the basis of spending time on their work. Time based wage is a remuneration system calculated based on the time contributed by workers. It is usually calculated based on hours worked. Thus, the total wage paid to each individual is equal to the total hours worked times the hourly rate. Thus, the higher the total hours worked, the greater the compensation received by the employee, assuming a fixed hourly rate. In another variation, rates may be based on daily rather than hourly.

Cash wages

Cash wages are compensation for employees that come in the form of spendable money. Cash wages can include actual cash currency, checks, and money orders. This type of compensation excludes benefits like health insurance, 401(k) contributions, and stock compensation. KEY TAKEAWAYS Cash wages are any pay or compensation that comes in the

form of spendable currency or other money. Cash wages normally make up the bulk of employment compensation for most workers, and are generally taxable. Non-cash compensation may sometimes also be offered or even preferred by employers, employees, or both, for various economic, tax, or business reasons.

Contract wages

A contract employee's salary is the amount of money they receive for performing work for their clients on a freelance basis. Specific contract employees' salaries vary based on the industry they work in, their experience and how much they charge their clients for their services. Contractors typically set their rates in one of the following formats: By the hour: Contractors who work intermittently for clients or complete small assignments for them often charge by the hour. By the project: Contractors often charge clients a flat rate to complete a large-scale project. By designated period: Contractors who work on an extended temporary basis, such as for a 30-day period, may charge by the week or month.

Advantages of wages

◇ More Work Flexibility: Employees working based on wages can work based on their selected hours. Further, since wages are earned based on hours worked, working overtime can help you earn more

◇ Less Work Responsibility: There are fewer obligations and work responsibilities for workers working on wages. Since work is done based on hours, employers assign work that's feasible to complete within those hours

◇ No Binding Contracts: Legally, no contracts can bind a worker on wages to stay with the firm. The individual is allowed to change jobs as they, please.

◇ Consistent Pay Calculation: Fair compensation is given to employees. Further, the accounting department in the company also benefits because of the uniformity of pay calculation where they are better able to issue everyone cheques a regular intervals.

◇ Compensated for Responsibilities: Since salaried employees have more responsibilities where they might have to work extra hours on unofficial times or weekends, they are compensated for that with bonuses or other perks.

◇ Employee Benefits: A salaried employee can enjoy all the employee perks like paid leaves, bonuses, insurance, remote working, and a flexible work schedule. Some companies even cover commutation and accommodation costs.

Incentives

In business, the objective of incentive is to increase employee Incentive in simple terms is something that encourages a person or organization to do or achieve something. It is

something that incites or has a tendency to incite a determination. This is usually given in cash or in kind.

Productivity, improve industrial and interpersonal relations, and as result increase the overall profit of the organization.

Types of incentives

- Financial incentives
- Non financial incentives
- skills incentives
- Goal incentives
- team work incentives

Financial incentives

Financial incentive pertains to those incentives which are in the form of money or can be measured in monetary terms. This is sometimes referred to as monetary benefit offered to consumers, employees, and organizations to encourage behaviour or actions which otherwise would not take place.

These incentives can be given on an individual or group basis and satisfy the monetary and future security needs of individuals. It lifts the eagerness and self-confidence of the employees thus, resulting in better productivity and performance.

Nonfinancial incentives

Non-financial incentives refer to incentives or rewards which cannot be measured in terms of money. All the needs of individuals are not satisfied by money only. Psychological, social, and emotional factors also play an important role in providing motivation. Some of the common non-financial incentives are:

Status: Status refers to reducing the position and prestige associated with a job in the organization.
Higher status helps to satisfy safety and social need, esteem needs, and self-actualization needs. Management provides status symbol facilities to its employees to motivate them. For example, costly furniture, well-furnished AC cabins, a car, etc.

Organizational climate: It refers to the characteristics which describe an organization and distinguish one organization from the other. Characteristics include individual autonomy, reward orientation, open communication, etc., which directly influence individual behaviour in a positive direction. Management helps to create an organizational climate and tries to motivate employees in this healthy working organizational climate.

Skills incentives

Skill-based incentives are rewards given to employees for reaching certain milestones or achieving a level of proficiency in a specific skill.

For example, an employee may be rewarded with a bonus once they have achieved a certain level of proficiency in software development. Or a sales team might be rewarded for reaching a certain number of sales in a given month.

These types of incentives can help to encourage employees to develop their skills by giving them tangible rewards for reaching specific milestones.

Goal incentives

Incentives are rewards or benefits given to employees for achieving specific goals or milestones. Incentives can take many forms, including cash bonuses, stock options, paid time off, and non-monetary rewards such as recognition and career advancement opportunities.

Team work incentives

Profit sharing is a team-based incentive plan in which you pay your employees a percentage of your company's overall profits. Profit sharing builds a sense of ownership among employees and encourages greater team performance levels. Staffers know that the better their performance, the better the business's financial picture, and the higher their own potential cash rewards.

Gain Sharing

Similar to a profit-sharing plan, gain sharing is a team incentive in which you reward employee groups for

measurable, non-financial achievements in pre-established areas. For example, teams may enjoy a bonus if customer satisfaction levels rise a certain percentage above figures from the previous year. The focus of this type of incentive is for employees to recognize the role they play in continually moving your small business forward in key areas.

Advantages of incentives

1.Improving business goals

One major benefit to an incentive program is that it encourages better performance in service of the company's stated business goals. Team members who feel proper motivation may be more likely to put extra time and effort into their work. For example, in sales, company leaders can create an incentive program specifically to encourage team members toward a certain sales goal. They might offer a $100 bonus for every member of a sales team upon reaching a company sales goal of $40,000 in a month.

1. Improving quality of work

Incentives can help to improve the quality of the work that a team accomplishes. For instance, an organization could implement a quality review for select projects and reward individuals who score well. Some incentives encourage fast work instead of quality work. Implementing a quality-based incentive process may be a useful way to balance out the

incentive program's effectiveness while improving the overall success of the organization.

2. Increasing morale

An incentive program can help individuals to feel appreciated at work because it shows that their manager noticed the hard work that they've been putting into their job. Praise and incentives can help improve team members' morale in the workplace. Happier team members may be more productive and more likely to pass along praise and encouragement to other team members. This means that everyone on the team may feel better appreciated for their work, which can have a major positive impact on the morale of the office overall.

3. Improving motivation

Incentive programs can help increase motivation for two reasons. First, the reward itself is an extra reason to work hard and strive for greater achievement. If a team member is finding it challenging to remain focused, having a specific, attainable incentive can be an excellent way to help them restore their motivation.

4. Encouraging teamwork

Incentives can be an excellent way to encourage teamwork and camaraderie. This may be especially true when management creates a group incentive that requires everyone on the team to succeed for anyone to earn the reward. One example would be when every member of a

sales team earns a bonus when the department reaches a certain quota of interested clients. There are many highly individualistic elements of compensation, like salary or benefits, so it can be useful for managers to be aware of the reward-based strategies that can encourage teamwork.

Etc.

CHAPTER-8

PROCESS PLANNING

INTRODUCTION

Process planning is how manufacturers determine the sequence of operations that they'll take to produce a part or make an assembly leading to the finished goods. When working in a job shop that produces one-of-a-kind products or where the same product is mass-produced, process planning is incredibly important.

But all manufacturers benefit from process planning. That's because process planning is basically the bridge between product design and product manufacturing. The purpose of process planning is to develop the most efficient sequence of operations. This includes having the right equipment and tools and meeting product requirements and specifications.

All of this is done with the goal of reducing costs and increasing productivity. Manufacturers have two ways to achieve these goals; one is manual process

Steps involve in process planning

Step 1: Develop Objectives

When beginning the planning process it is important to first develop objectives. These objectives could vary in nature

depending on the goals or mission of an organization or customer or consumer needs. Nonetheless, it is crucial to start with the high-level objectives of what the organization is trying to achieve. For example, in a strategic plan, an objective might be to increase staff retention by 10%. In an operational plan, an objective might be to decrease waste by 20%. Note that important common factors include a metric by which the objective will be measured and a specific area of focus.

Step 2: Develop Tasks to Meet Objectives

Once the objectives have been determined, tasks need to be developed to help achieve those objectives. Objectives are often large, important goals that need to be accomplished over an extended period of time. Tasks, on the other hand, are smaller goals that one needs to accomplish in order to complete the larger goals.

Step 3: Determine Needed Resources

Resources in any plan can be more than just physical objects or equipment. Resources can also be the people needed to complete tasks and achieve goals. It may seem overwhelming when determining the needed resources for your plan, especially if the plan is large and complex. One way to determine needed resources is to go back to the objectives of the plan and more specifically the tasks and determine what resources are needed to complete the objective or task.

And many more.

Selection of machine tool

Machine tool selection is an important decision-making process for many manufacturing companies. Improperly selected machines degrade the overall performance of a production system. The speed, quality, cost, and capacity of manufacturing strongly depend on the type of the machine tool used Production capacity, cost of the equipment and its ability to give satisfactory service are major factors to be considered when making a selection of machine tools

The process plan

Process plan is a preparatory step before manufacturing, which determines the sequence of operations or processes needed to produce a part or an assembly. This step is more important in job shops, where one-of-a-kind products are made or the same product is made infrequently.

Types of process planning

1 Manually process planning

Also known as man-variant process planning, this is the most commonly used type of process planning. A process planner will select the combination of processes needed to produce the finished part. These are engineering drawings that define the part requirements and the availability and capabilities of machines and processes. The selection leads to criteria, such as production costs or time, machine utilization and routine

will also impact the plan. Usually, the criteria are based on a family of parts rather than the processes and machines needed to produce a single part.

2 Computer-Aided Process Planning

This is a more efficient type of process planning, less tedious and less likely to make mistakes. Automating process planning systems removes the human element in the steps between preparing engineering drawings and finishing the process plan for all manufacturing operations. There still must be human oversight to ensure compatibility with the available processes. Whether to turn, mill or stamp is determined by the product requirements, quantities ordered and capabilities of the processes. This still reduces process planning time, and the skills required of a process planner and makes for more consistent and accurate plans that increase productivity and reduce costs.

3 Generative Process Planning

Generative process planning takes in-process information and uses it to automatically create a process plan for a new component. This information is pulled from manufacturing databases and doesn't involve much human oversight. After reviewing the design, the operation sequence is automatically produced. Of course, the software must be encoded with knowledge of manufacturing. In fact, machine selection, tool selection, process optimization and more can also be automated by this system. There are hurdles to clear,

however, such as transforming component data and decision rules into the system.

The work place layout

The layout and design of a workspace will determine the flow of people, materials, and equipment. Layout planning is deciding on the best and most efficient arrangement of all physical resources that will need space in a work area or a facility

One of the best ways to avoid workplace strains and sprains is to design a workspace that reduces injury risk factors. Factors to consider include:

Height of the work to be performed: Workers should be able to sit or stand erect without having to lean forward. Storage should be organized such that the heaviest items are stored between knee and shoulder height to avoid bending and reaching overhead.

Standing workstations: Long-term standing can place excessive stress on the back and legs. Where long term standing is required, a footrest or rail, resilient floor mats, height-adjustable chairs or stools, and opportunities for workers to change positions should be provided.

Seated workstations: Chairs should be fully adjustable, especially where workstations are used by multiple users.

Overhead storage: If items must be stored overhead, a warehouse ladder, steps tool or other means should be provided to achieve better lifting conditions

Storage space requirements

Storage space should be adequate, considering the material requirements and the load bearing capacity of the soil if heavy materials are to be stored. Aisles are passages for the movements of men and materials. Depending upon the type and size of the material to be stored, the amount and frequency of men movements and other traffic involved, and the size of material handling equipment, the aisles and the sub-aisles are designed. Generally aisles can be 1.5-3 metres wide and sub-aisles between racks may be 75 cm wide. The aisles should not have many turns or blind corners which may cause accidents.

A window as a part of the enclosed store space can be used for issuing and portable products, which can be passed on to the workers through the same so that operators can be kept outside the store room. It is very necessary for an effective store room

Design of workplace layout.

The relationship between workplace design and wellness has become increasingly emphasized by employers as an important factor of worker wellness. In industrial settings,

workplaces must be designed to ensure workers can safely perform routine functions and efficiently conduct emergency safety procedures. In an office workplace, employers increasingly focus on designing environments that optimize employee wellness by maximizing conditions such as air quality, lighting, and ergonomics.

Workplace design principles involve efforts to optimize the safety and health conditions of regular work activities through measures such as ergonomic seating and temperature control, as well as efforts to protect workers in high-risk industries through measures like designing safe navigation routes through construction sites. Some workplaces use detailed aesthetic designs to reduce worker stress. These might involve the use of plants to improve mood, the use of specific types of lighting, and the use of alterations to make a space “feel” less crowded.

Factor considered in designing a workplace

A well-designed office space should include private zones where employees can focus on the tasks assigned to them without interference from colleagues. You can also consider allocating spaces that promote rest and relaxation so employees can manage demanding situations better.

Ergonomics: A good office design ensures that employees can work without discomfort or pain. On the other hand,

disorganized and cluttered space leads to wasted time looking for supplies while also contributing to feelings of stress and frustration — all things which can contribute to physical pain.

Workflow: Whether horseshoe-shaped, cluster, a straight line, circular or other setups, the office floorplan needs to make the flows of work as seamless and efficient as possible.

Maximization: Space can be costly, especially when talking about real estate in a place like downtown NYC. As such, all real estate in the office should be optimized while still giving enough space to promote the well-being and efficiency of the staff.

Unrestricted Movement: Clutter and cramped conditions in the office may reduce efficiency. These problems can be remedied by better organization and by providing enough room for unrestricted movement in the office floor plan. Also, the furniture and equipment should be positioned to allow better access to cleaning, maintenance and egress in cases of emergency. Technological requirements should be considered here too.

Allocation: When divvying up real estate to different departments, the degree of relationship of one department to another is taken into account.

Uniformity: Furniture, equipment and tooling used in an office should be uniform to guarantee flexibility and a pleasing, calming appearance.

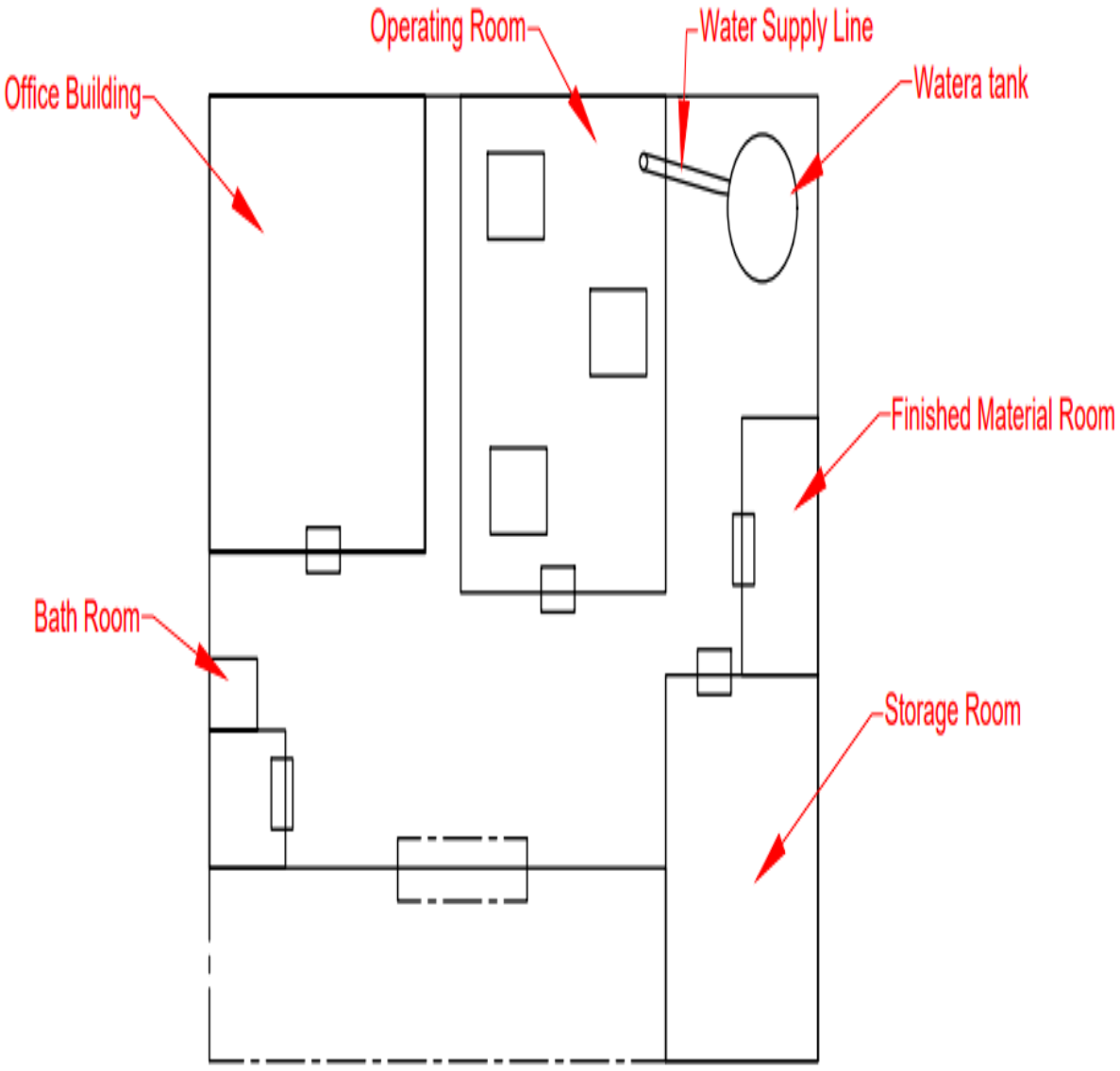
Compatibility: Office systems and procedures must be compatible with the office layout for employees to be efficient. To attain this, make sure employees and equipment are positioned to fit and make logical sense when used with the procedures and office systems in operation.

Safety: Safety in the office is a very crucial aspect. Fire exits, emergency exits and fire alarms should be provided and clearly marked to ensure the safety of staff and visitors.

CONCLUSION

We have decided that our workplace layout will look like this.

Workplace Layout



CHAPTER-9

COST ANALYSIS

INTODUCTION

Cost analysis is the process of examining and understanding all the expenses incurred by a business. It involves breaking down and evaluating costs related to production, operations, and other activities. This helps businesses make informed decisions, manage resources efficiently, and optimize financial strategies to enhance overall profitability. The total cost is divided into different headings known as "Elements of Cost".

Total cost of a product can be divided into three main "Elements".

These are:

1. Materials.
2. Labor.
3. Expense.

1. Materials Cost

These can be further classified into:

(1) Direct materials. Direct materials are the substances used in factories that, when worked on during production, take the final shape of the main product or a part of it. They're often called "Productive Materials."

(2) Indirect materials. Indirect materials are necessary substances in factories that support the transformation of direct materials into the final product. They are essential in different areas of the production process but don't directly become the end product.

CALCULATION OF MATERIAL COST

For the calculation of material cost following procedure should be adopted:

1. **Measure Components:**
   - Find the size of each part using measurements, considering extra material for machining.
2. **Total Product Volume:**
   - Add up the volumes of all parts to get the overall product volume.
3. **Calculate Material Weight:**
   - Multiply the total volume by the material density to find the material weight.
4. **Determine Material Cost:**
   - Multiply the cost per unit weight by the total material weight to get the material cost.

2. Labour Cost

Labour employed in any factory may be of two classes: (i) Direct labour, and (ii) Indirect labour.

(i) Direct Labour. Direct Labour refers to the workers who are directly involved in manually processing materials or using machines. These workers operate production machines in various areas like machine shops, welding shops, pattern making shops, electric winding shops, and assembly shops.

(ii) Indirect Labour. Any other labour, which helps the productive labour in performing their duties is known as "Indirect Labour".

CALCULATION OF DIRECT LABOUR COST

To calculate direct labor cost, the estimator needs to understand all production operations, tools, and machines involved. Consulting the production department for accurate time estimates for each operation is crucial. This ensures an informed and precise calculation of labor costs.

3. Overhead

Expenses in a factory include various costs like advertising, rent, plant depreciation, packing, transportation, and sales-related payments. Essentially, any cost apart from direct materials and direct labor, including indirect materials and indirect labor, is categorized as "Expenses."

Expenses may be of two classes:

(i) Direct for chargeable expenses,
and (ii) Indirect Expenses.

(i) Direct Expenses. Direct Expenses are specific costs directly related to a particular job. Examples include expenses for special tools (like jigs and fixtures), unique patterns, or experimental work dedicated to that specific job.

(ii) Indirect Expenses. These are also known as overhead charges, on-costs, burden, indirect charges, indirect costs, secondary costs or supplementary costs. These can be further classified as:

(a) Factory expenses.
(b) Administrative expenses.
(c) Selling expenses.
(d) Distribution expenses.

(a) Factory Expenses. Factory Expenses, also known as Works Ongoing Costs, encompass all the spending directly involved in producing a product within the plant. This includes costs for things like indirect materials and indirect labor, essential for the overall production process.

(b) Administrative Expenses. Administrative Expenses cover the costs associated with general office operations, such as salaries for office and executive staff, telegraph and telephone charges, and the depreciation of office buildings and equipment. These are also referred to as

"establishment on cost" or "office expenses," reflecting the administrative aspects of running the business.

(c) Selling Expenses. These overheads include all the expenditure made on the salaries of persons working on sales department, advertising expenses, agency expenses etc.

(d) Distribution Expenses. These overheads include all the expenses made on holding finished stock, dispatching them to the customer, packing cost etc.

FIXED AND VARIABLE OVERHEADS

All the overheads described above can be classified into following two forms:

(i) Fixed overheads.

(i) Variable overheads.

1. Fixed Overheads. These are those indirect expenses, which remain constant whatever may be the volume of production.

Examples of these overheads are:

(a) Salaries of Staff.
(b) Depreciation of machines and equipment.
(c) Interest on capital invested.
(d) Rent of building and insurance.

2. Variable overheads. These are those indirect expenses, which vary with volume of production. Examples of these overheads are:

(a) Power or fuel consumed.
(b) Consumable store supplies.
(c) Repairs and maintenance. Expenses on tools

COMPONENTS OF COST

The various components of cost are:

1. Prime cost.
2. Factory cost.
3. Office cost.
4. Total cost.

1. Prime Cost. It consists of direct material cost, direct labour cost and direct expenses.

i.e. Prime cost = Direct material cost + Direct labour cost + Direct expenses.

Prime cost is also named as "Direct Cost'.

2. Factory Cost. It consists of prime cost and factory expenses.

i.e. Factory cost Prime cost + Factory expenses.

Factory cost is also named as "Works Cost".

3. Office Cost. It consists of factory cost and administrative expenses.

i.e. Office cost = Factory cost + administrative expenses. Office cost is also named as manufacturing cost or cost of production.

4. Total Cost. It includes office cost and selling and distribution expenses.

i.e. Total cost Office cost + Selling expenses + Distribution expenses.

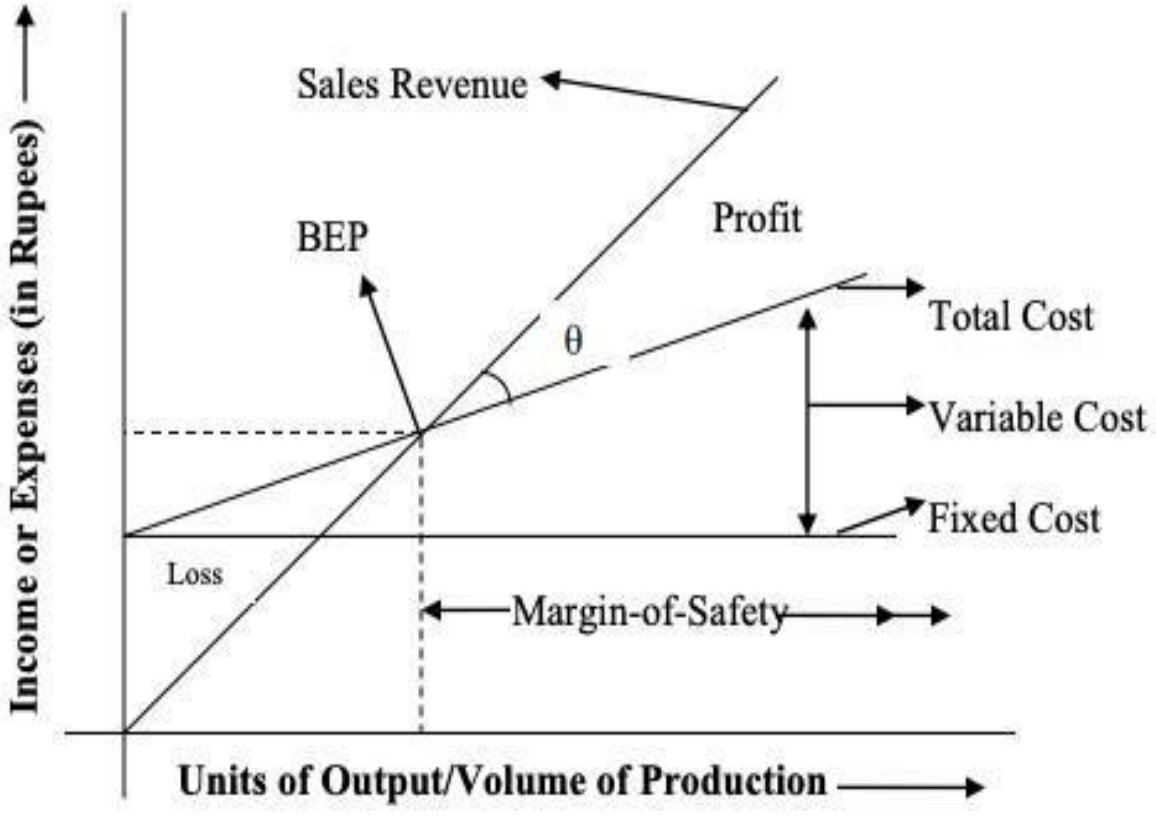
DEPRECIATION

Depreciation is the decrease in efficiency and value of a machine or asset over time due to use. To prepare for replacing it when it becomes outdated, a portion of profits is set aside annually. This ensures funds are available for upgrading equipment when needed.

BREAK-EVEN ANALYSIS

Break-even analysis shows when a business neither makes a profit nor incurs a loss. It pinpoints the point where total revenue equals total cost, known as the Break-even Point (BEP). This balance between costs and revenue helps businesses understand at what level they cover their expenses without making a profit or facing a loss.

Break-even Chart (BEC):- A Break-even Chart (BEC) is a helpful tool for solving industrial problems. It plots production volume against costs and income. The angle of incidence, formed by the income and total cost lines, determines the break-even point—where earnings cover expenses. Below this point, the company incurs losses; beyond it, profits are made. The Break-even Point (BEP) relies on fixed and variable costs, making the BEC a visual representation of the cost-revenue relationship at a specific time.



CHAPTER 10

PRODUCT DESIGN AND ASSEMBLY DRAWING

THEORY OF "MANUAL HYDRAULIC JACK WITH RETRACTABLE HANDLE"

INTRODUCTION

A jack is a mechanical lifting device used to apply great forces or lift heavy loads. A mechanical jack employs a screw thread for lifting heavy equipment. A hydraulic jack uses hydraulic power. The most common form is a car jack, floor jack and garage jack, which lifts vehicle so that maintenance can be performed. Jacks are usually rated for a maximum lifting capacity (for example 2Tons or 4Tons).

A hydraulic jack is a device that uses the power of liquid to lift heavy objects. It works by using a piston and cylinder filled with hydraulic fluid. When you apply force to the jack's handle, the fluid gets compressed, causing the piston to

move and lift the load. It's a handy tool for lifting cars or heavy equipment!

A bar is inserted into the handle socket of the jack, when pumped, the hydraulic fluid forces by a piston and go through a one-way valve. The fluid force acts at the bottom of the bigger cylinder, allowing the ram to go upwards. The other valve is blocked so pressurized fluid can't back until it is unscrewed. This is how the hydraulic jack works.

THE MANUAL HYDRAULIC JACK WITH RETRACTABLE HANDLE CONSISTS OF THE FOLLOWING MAIN COMPONENTS:-

1. Torsion spring
2. Torsion bar
3. Bolt
4. Housing
5. Hub
6. Gear

1.Torsion spring:- A torsion spring is a spring that works by twisting its end along its axis, that is a flexible elastic object that stores mechanical energy when it is twisted.



FIG.1. TORSION SPRING

2. Torsion bar:- A torsion bar is a long spring element with one end held rigidly to the frame and the other end may be twisted by a lever or connected to an axle or left freely suspended.

3. Torsion bar holder:- It is a metal rod or pin for fastening objects together that usually has a head at one end and a screw thread at the other and is secured by a nut.

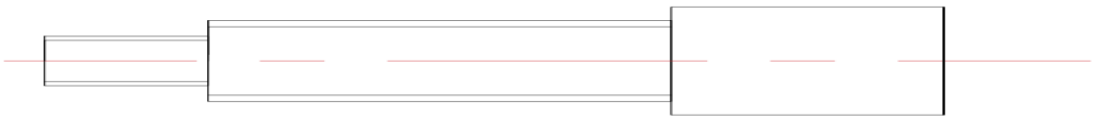
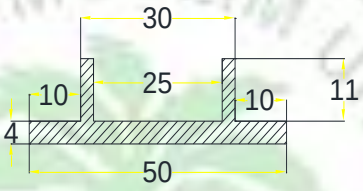


FIG.3. BOLT

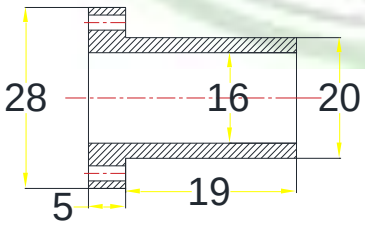
4.Gears:- They are mechanical components used for transmission of power from one place to another. They are of various types .



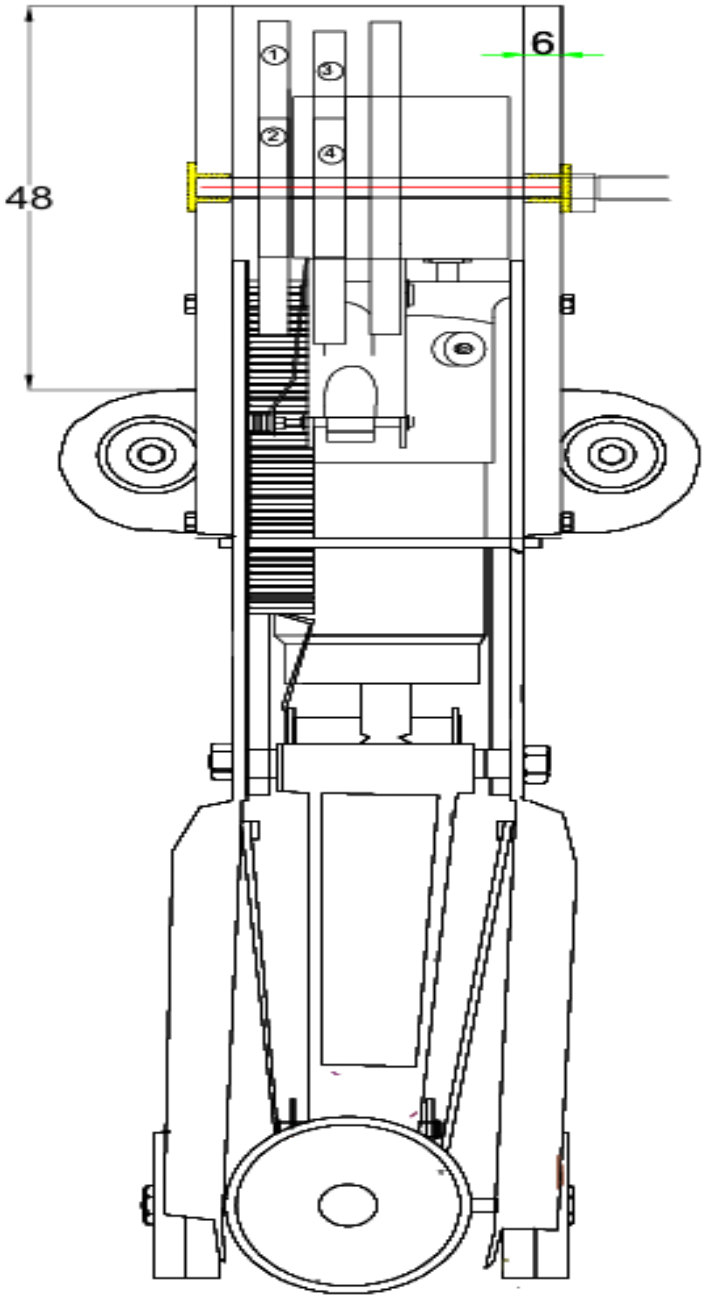
Housing:- It is a mechanical part used to hold the various components altogether.



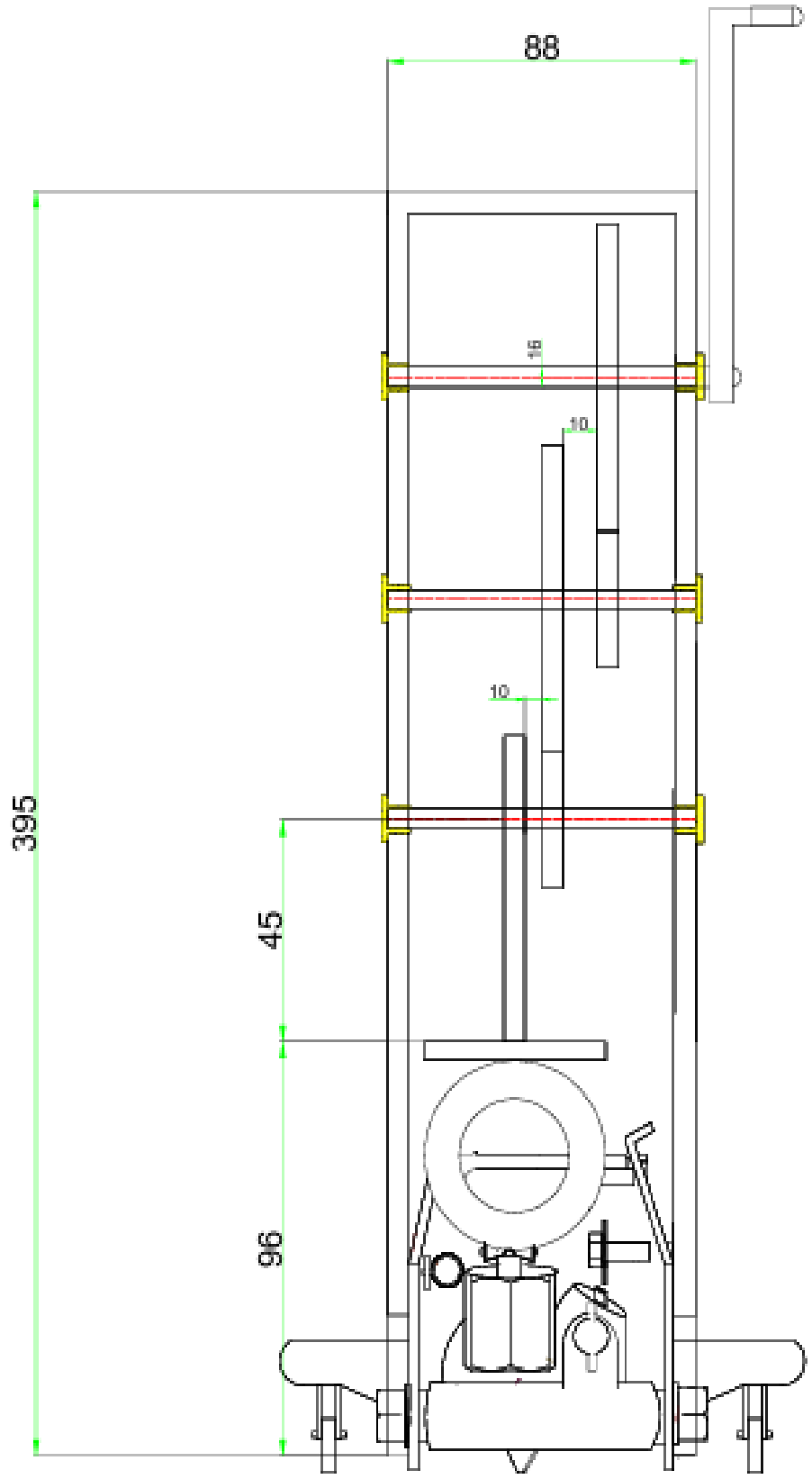
Hub:- It is a mechanical part used to fit in the gear for achieving proper rotation



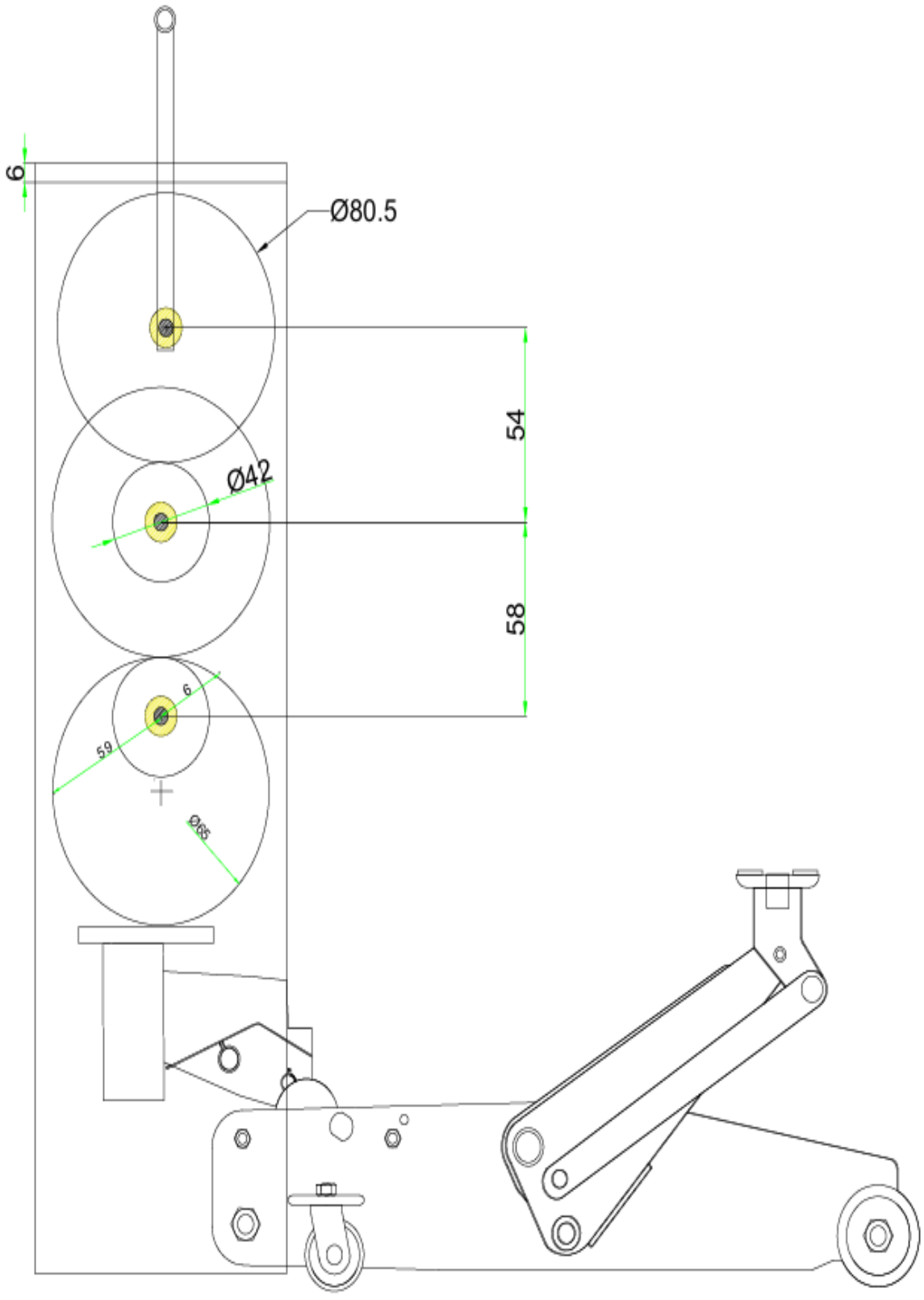
ASSEMBLY DRAWING



PLAN



ELEVATION



SIDE VIEW

A brief description of all the parts used with their function is listed below:-

PARTS LIST

PART.NO	DESCRIPTION	FUNCTION
1	Torsion spring	Store and release energy.
2	Torsion bar	To support the spring
3	Bolt	To hold the torsion bar.
4	Gears	To transmit power from one point to another
5	Hub	Attached to a shaft for
6	Housing	For smooth engagement of shaft to frame.
7	Shaft	To support gears and cams.
	Shaft steps	To support bearings and housing
9	Cam	To reciprocate follower
10	Steel frames	To support components.

DESIGN OF THE MANUAL HYDRAULIC JACK WITH RETRACTABLE HANDLE

S.NO	MATERIAL	DENSITY(gm/cm^3)
1.	Mild steel	7.85
2.	Carbon steel	7.8

3.	Cast iron	7.13

WEIGHT IF RAW MATERIAL:

For mild steel rod of 16mm × 25mm

$\{\pi \times [0.8]^2 \times 2.5cm$

$V=5\ cm^3$

W=5 x 7.85

=39.25 gm

Estimating the weight of torsion spring-

Material: carbon steel

Dimensions:-

Thickness: 3mm

Length of leg 1:- 52.64mm

Length of leg 2:- 50.00mm

$V = \{\pi \times [0.15] \times 33$

$=9cm^3$

Weight : -v x d

=70gm

Estimating the weight of torsion bar:

Material: Mild steel

Shape:- Hollow cylindrical

Dimensions:-

Length:- 14mm

Internal diameter:- 8mm

External diameter:-14mm

Volume required:

$V_t = \{\pi \times [0.7]^2 \times 1.5\}\ cm + \{\pi \times [1.25]^2 \times 0.3\}\ cm$

$=3.77cm^3$

Weight of torsion bar = 3.77 x 7.85 = 29.5 gm

Estimating the weight of bolt:

Material:- Mild steel

Shape:- Cylinder with threads

Dimensions:-

Length:-33mm

Threading:- M9

Length:- 17mm

Threading:- M5.5

Length :- 6mm

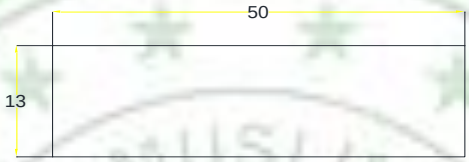
Weight:-19gm

Estimating the weight of housing:

Dimensions for raw material: 50 mm x 15 mm

$v = \pi r^2 h$

$=3.14 \text{ x } (2.5)^2 \text{ x } 1.5$



$=25.512 cm^3$

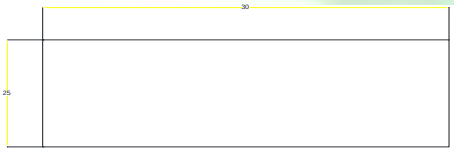
Weight of housing=25.512 x 7.85

=200.26 gm

Weight of 6 housing = 200.26 x 6

= 1201.56 gm

Estimating the weight of hub:



Dimensions for raw materials:30mm X 25mm

Volume of hub raw materials=

$\pi r^2 h$= 3.14 X $(1.5)^2$ X 2.5

=17.662 cm^3

Mass of hub= 17.662 X 7.85=138.64gm

Weight of 4 hubs= 138.64 X 4= 554.56gm

Estimating the weight of shaft step:

Dimension of raw materials: 16mm X 10mm

$$v = \pi r^2 h$$

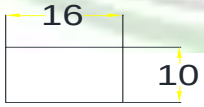
=3.14 X $(0.5)^2$X 1.6

=1.256cm^3

Weight of step = 1.256 X 7.85 =9.8596gm

Weight of 6 shaft steps= 9.8596 X 6

=59.157gm



Estimating the weight of shaft:

Dimensions of raw material: 76mm X 16mm

$$v = \pi r^2 h$$

=3.14 X $(0.8)^2$ X 7.6

=15.27cm^3

Weight of 1 shaft= 119.87gm

Weight of 3 shafts=359.61gm

Estimating the weight of shaft steps:

Dimensions of the shaft steps: 16mm X 10mm

$$v = \pi r^2 h$$

=3.14 X $(0.5)^2$X 1.6

=1.256cm^3

Weight of 1 shaft step=1.256 X 7.85

=9.86gm

Weight of 6 shaft steps=9.86 X 6

=59.16gm

Estimating the weight of Ring:

Dimensions of the ring:

Outer diameter=50mm

Inner diameter=26mm

Length=30mm

$$v = \pi r^2 h$$

=3.14X $(2.5)^2$X3

=58.875cm^3

=3.14 X$(1.3)^2$X3

=15.91cm^3

Net Volume =58.875-15.91=42.965cm^3

Weight of ring = 42.965 X 7.85=337.27gm

Estimating the weight of frames:

Dimensions of steel frames: 395mmX 45mm X 6mm

$$v = LX\ bXh$$

=39.5 X 4.5X0.6

=106.65cm^3

Weight of 1 steel frame= 106.65 X 7.85

=837.202gm

Weight of 2 steel frames =837.202 X 2

=1674.40gm

Dimensions of cross-Bars:140mm X 48mm X 6mm

$$V=L\ X\ b\ X\ h$$

= 14 X 4.8 X 0.6

=40.32cm^3

Weight of cross-bar :40.32 X 7.85

=316.512gm

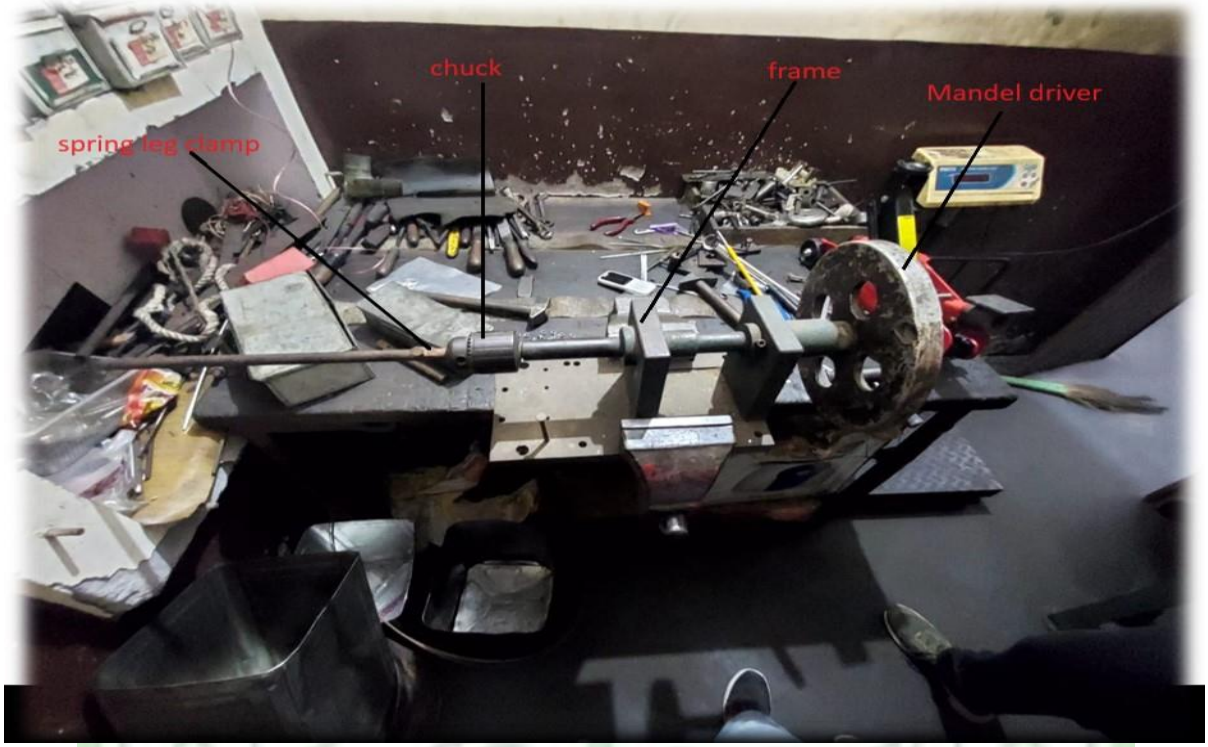
Total weight of steel frame= 1674.40+316.512

=1990.91gm

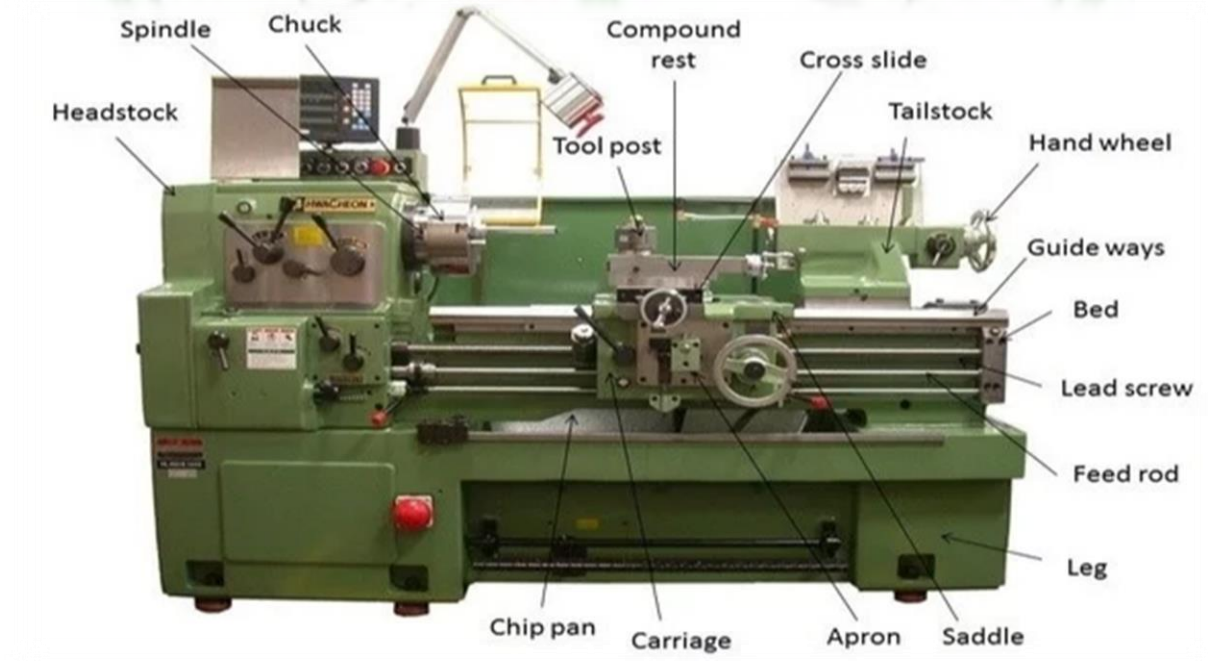
=2kg

MACHINES USED:-

TORSION SPRING MAKING MACHINE:-



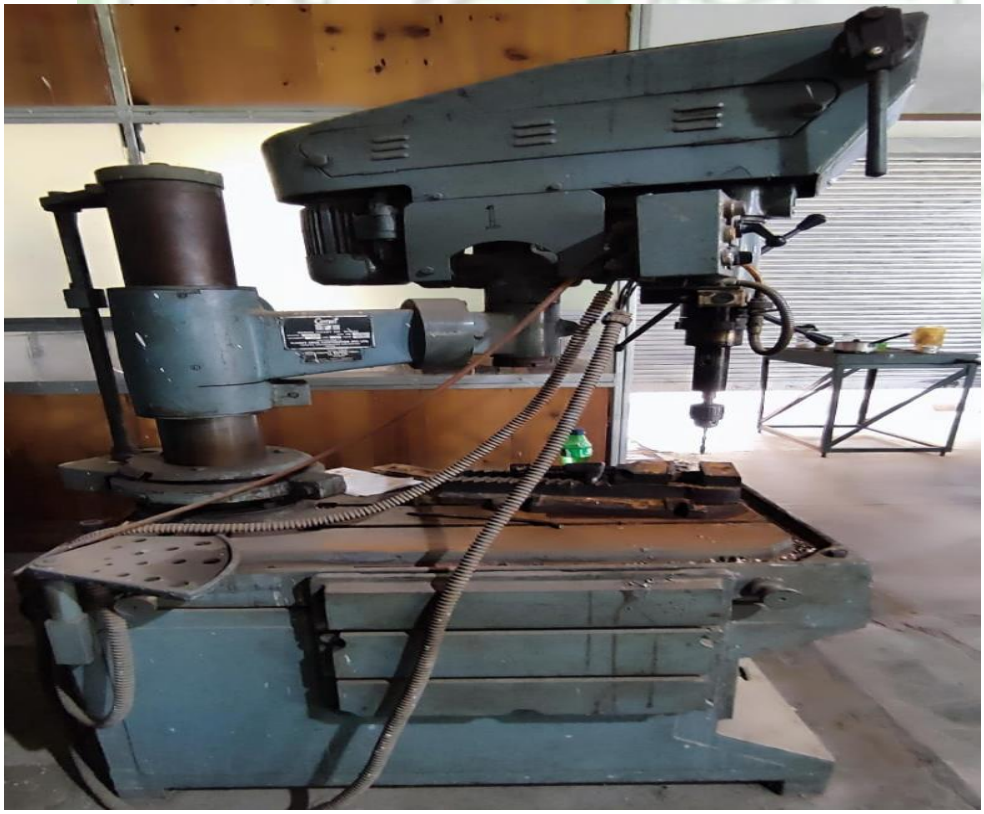
LATHE MACHINE :



CUTTER MACHINE:



DRILL-MACHINE



CHAPTER 11

RAW-MATERIALS

All the materials used in project is given and described in the table below.

S.NO	MATERIAL	PROPERTIES	AVAILABILITY	RATE
1	Mild steel	High tensile strength ,impact strength and good weldability	Easily available	75/kg
2	Stainless steel	Corrosion resistance ,high tensile strength ,very durable ,easily formability and lubrication	Sometimes not available	120/kg
3	Carbon steel	Strong and highly durable,	Easily available	90/kg

higher strength ,high and low temperature strength,

CHAPTER-12

FINANCIAL ESTIMATE

Introduction

A financial estimate is a term used to describe the prediction of finances or using some practical methods to approximate the outcomes or costing of product(s) in an industry. These are done on some value based available data or assumptions done in the past of industry. Hence, they are very important parameter for an industry.

Benefits of financial estimates are given below:-

1. They help in decision making process about the various important fields.
2. They control the flow of funds in an industry.
3. They aid in prevention of wastage of time, money or effort of any individual working in that industry.

Hence, there are two important pillars of financial estimating namely- Costing and Estimation. Each one has different aims which are discussed below.

Costing

Costing as defined by IFAC (International federation of Accountants) is-"*Costing is the process of calculating and allocating the cost of products, services or other expenditures incurred upon the object*". This includes various methods and techniques for expenses associated with the good or service produced.

There are many types of costs which has been costing -

1. Material cost
2. Labour cost
3. Expenses

Material cost

This is further subdivided into two categories –

- **Direct Material cost**
  In our project the direct materials are tabulated below with price –

S. No.	Parts for single unit	Number of parts required for single unit	Weight for single unit (in gms)	Parts for 700units	Weight for 700 units (in Kg)
1.	Hub	4	554.56	2800	388.192
2	Housing	6	1201.56	4200	841.092

3	shaft	3	359.61	2100	251.73
4	Ring	1	337.27	700	236.08
4	Shaft steps	6	59.16	4200	41.42
5	Frames and crossbar	3	2000	2100	1400

Considering the production of 700 units in one month
For one month raw material taken is shown below:-

Name of the material	Quantity of the material taken in kg	Cost of the material per kg	Total cost of the material
Mild steel rod	1758.514	60	105510
Mild steel flats	1400	60	84000
Carbon steel	49	90	4410

Total cost =₹193920

Name of the component	Cost of the component in ₹

Hydraulic Jack	1080
Gears	880
Bearings	60
TOTAL	2020

Total cost of 700 components in one month =₹1414000

Total material cost =₹1607920

- **Indirect material cost**

Name of the material	Quantity of material used	Cost of the material/month
Lubrication oil	5L	750
Coolant	60L	6000

Total indirect material cost = ₹ 6750

Labour cost

1. Direct labour cost

Salaries of operative staff

S No.	Designation	No. of operator	Salary /month	Total Salary
1.	Lathe operator	3	14875	44625
2.	Drill-machine operator	1	8330	8330

3.	Spring-maker/assembler	1	10000	10000

Total cost =₹62955

2. Indirect labour costs

Salaries of non-operative staff

S No.	Working Hours	Number	Designation	Salary/ month	Total Salary
1.	Full time	1	Accountant	15000	15000
3.	Full time	2	Helper	12000	24000
4.	Full time	1	Watchman	10000	10000
5.	Part time	1	Sweeper	6000	6000
6.	Full time	1	Director/supervisor	25000	25000

Hence, total cost of non-operative staff in a month is ₹80000

Total cost of staff = ₹33205+₹68000=₹101205/month

Cost of machines and tools

S No.	Capital item	Qty.	Cost /item	Total cost	Life (yr.)	Salvage value	Depreciation Per month
1.	Drill machine	1	35000	35000	20	1575	292
2.	Lathe Machine	3	125000	375000	20	2500	3120

3.	Spring Machine	1	8000	8000	20	360	66
4.	Cutter	1	15875	15875	20	2500	668.75

Total cost of tools and machines is ₹168000

Capital cost

Considering a constructed factory shed at rent for ₹10000 per month

S no.	Capital Item	Qty. (m^2)	Expenditure/m^2	Total cost	life	Depreciation /month
1.	Land	150	-	-	-	-
2.	Office	30	1100	33000	80	275
3.	Finished material Room	20	260	5200	80	43
4.	Washroom	5	120	600	80	5
5.	Drinking Water	5	80	400	10	3
6.	Store room	30	215	6450	80	54
7.	Furniture	-	3000	3000	20	25
8.	Water tank charges	1500L	1500	1500	10	14

Hence total Expenditure cost=₹50150

Indirect Expenses and Overheads

S no.	Name of items	Cost of items needed in ₹ per month	Quantity
1.	Stationery	1000	-
2.	Soap	200	10
3.	First Aid box	300	-
4.	Diesel	4000	40L

Total Indirect expenses cost =₹5500

Overhead expenses

Name of the material	Cost of the material/month
Electricity charges	5000
Cost of transportation	1200
Telephone charges	1500
Loading/unloading cost	1000
Total depreciation	4146
Rent	10000

Total cost of overheads = ₹22846

Calculation of the cost of Product

Direct material cost= ₹1607920

Loan / month = ₹8475

Insurance paid per month =₹5000

Indirect Material cost = ₹6750

Total Labour cost= ₹101205

Indirect expenses =₹5500

Overheads = ₹22846

Hence total cost = ₹1757696

Cost of single unit = ₹2510

Taking profit of 15% as profit, we get

Selling price = 2510+ 15% of 2510=2510+195= ₹2885

Selling price of single unit = ₹ 2885

CHAPTER-13

BREAK EVEN ANALYSIS

Introduction

Break even analysis is a study of the relationship between the factors affecting profits. It indicates the the level the cost of product and revenue are in equilibrium. It directly relates to total revenue and total cost .During this analysis there exists a point which tells us at what level of sales and production the bussiness can cover its expenses. Thus, resulting in neither loss nor profit. Such a point is called as "**break-even point or BEP**". Beyond it the compnany starts generating profit while below it compnany orperates at loss.

Hence, this type of analysis is useful to answer the following questions mentionrd below-

1. What volume of sales will be necessary for covering returns or capital employed.
2. To find the price of an article to give a desired profit.
3. To determine variable cost per unit.
4. To estimate the future scale for production and revenue.

Break-even chart(BEC)

BEC is very useful means for solving the industrial problems. Here income (plotted on ordinate) and production volume (plotted on abssicca) are represented. The income line makes a specific angle called as "*angle of incidence*" and represented as "θ". The point where the income curve cuts the Sales revenue or production volume is called the "Break – even point".

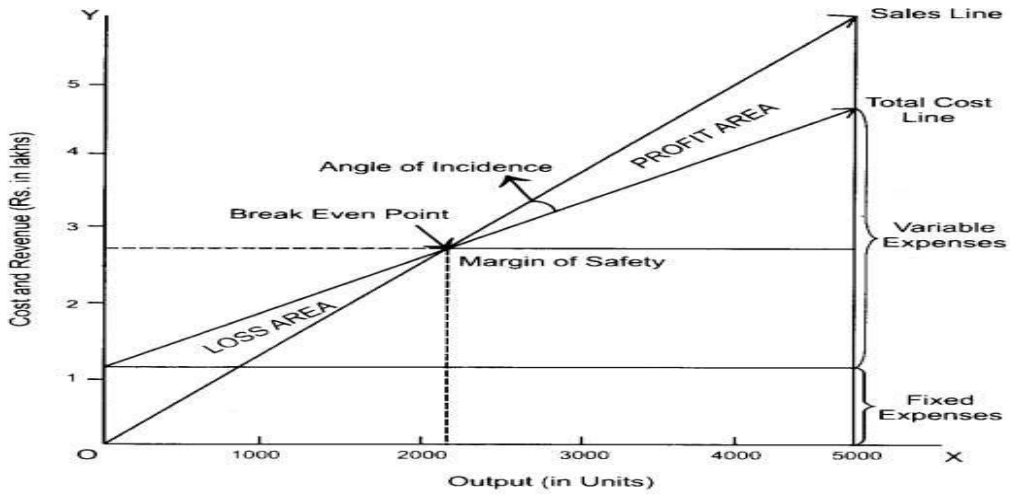
BEC depends upon the fixed cost and variable cost.

Fixed cost- The cost which the manufacturer has to pay in fixed amount and need to be given at any condition. Eg-

Electicity bill, government taxes etc.

Variable cost-The cost which fluctuates according to the need of item , demand and time . Eg- Raw materials , storage cost etc.

The graphical representation of this chart is given below-



Calculation of Break-even point

Let break-even point be denoted by BEP(in units) ---Quantity produced or sold

Variable cost is denoted by VC(in rupees/Unit)

Fixed cost is denoted by FC(in rupees/Unit)

Selling price is denoted by SP(in rupees /Unit)

Therefore, for calculation of BEP,

BEP=FC/SP-VC

Here,

FC=Fixed cost = ₹128826

SP=Selling Price =₹2885

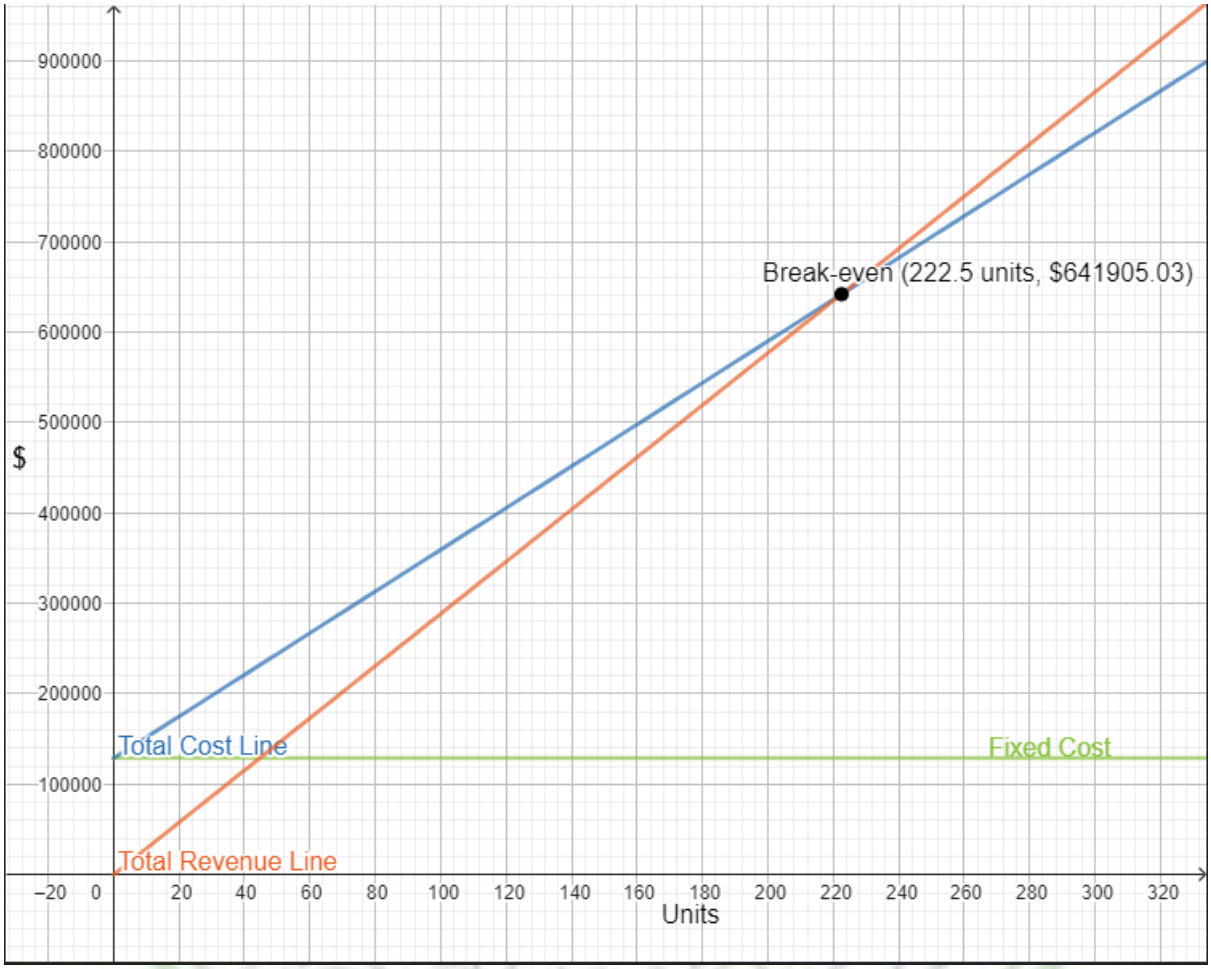
VC= Variable cost =₹2306

BEP=FC/SP-VC=223 units

BEP in rupees = 223 X 2885=₹640470

Total sales should be greater than or equal to ₹640470

Conclusion



We concluded that for a successful hydraulic jack with retractable handle the minimum sales should be of 223 units and with sales of ₹ 640470.

This type of arrangement is easy to manufacture , reduces human effort and is affordable .

CHAPTER-14

DISCUSSION AND CONCLUSION

DISCUSSION:-

In conclusion, manual hydraulic jacks provide a practical and efficient means of

Lifting heavy loads using hydraulic pressure generated by manual force. Their simplicity, reliability, and versatility make them valuable tools in various applications, from automotive repairs to industrial settings. Regular maintenance and proper usage are crucial for ensuring optimal performance and longevity.

COCLUSION:-

The manual hydraulic jack with a retractable handle offers a convenient and efficient solution for lifting applications. Its retractable handle enhances portability and storage, making it a versatile tool for various situations. The hydraulic mechanism provides reliable lifting power, making it suitable for tasks such as lifting vehicles or heavy equipment. Regular maintenance and proper usage are crucial for ensuring the longevity and effectiveness of the hydraulic jack.